

**A PRIVACY-PRESERVING FEDERATED
ENSEMBLE XGBOOST FRAMEWORK FOR
PROACTIVE PATIENT READMISSION
MANAGEMENT**

NAYAN JIKAR



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING
FACULTY OF ENGINEERING AND TECHNOLOGY,
DATTA MEGHE INSTITUTE OF HIGHER EDUCATION AND
RESEARCH (DU)
SAWANGI (MEGHE), WARDHA, MAHARASHTRA-442107
FEBRUARY, 2026

**A Privacy-Preserving Federated Ensemble
XGBoost Framework for Proactive Patient
Readmission Management**

The Major Project-II report submitted in partial fulfilment of the
requirement

For the Award of the Degree of

Bachelor of Technology

by

Nayan Jikar

(Enrollment No:- R-16654)

Under the Guidance of

Dr. Asadullah



Department of Artificial Intelligence and Machine Learning
Faculty of Engineering And Technology,
Datta Meghe Institute of Higher Education and Research (DU)
Sawangi (Meghe), Wardha, Maharashtra-442107

February, 2026

Major Project-II Approval

Major Project-II entitled: **A Privacy-Preserving Federated Ensemble XG-Boost Framework for Proactive Patient Readmission Management** by **Nayan Jikar** is approved for the degree of Bachelor of Technology (B. Tech.) in **Artificial Intelligence and Machine Learning** of Faculty of Engineering And Technology, Wardha.

Examiners

(i) _____

(ii) _____

Dr. Asadullah

Supervisor

Dr. Utkarsha S Pacharaney

HOD/ Dean Academics, FEAT

Prof. (Dr.) K.T.V. Reddy

Dean, FEAT

Date: _____

Place: Wardha

Declaration

I declare that this submission is my own work and ideas. If I have used someone else's words or ideas, I have properly cited and referenced them.

I confirm that I have followed all rules of academic honesty and integrity. I have not misrepresented, fabricated, or falsified any information, data, or sources in this submission.

I understand that breaking these rules can lead to disciplinary action by the institute and possible legal consequences from the original sources if proper credit or permission has not been given.

(Nayan Jikar)

(Enrollment No:- R-16654)

Date: _____

Place: Wardha

Acknowledgment

The making of the project needed the cooperation and guidance of a number of people. I therefore consider it my prime duty to thank all those who had helped me through this venture. It is my immense pleasure to express my gratitude to **Dr. Asadullah** as a guide who had provided me with constructive and positive feedback during this major project-II.

I sincerely thank the Head of Department and Dean Academics, **Dr. Utkarsha Pacharaney**, and all other staff members of the Artificial Intelligence and Machine Learning department for their kind co-operation.

I would like to thank **Prof. (Dr.) K.T.V. Reddy**, dean of our institution, for providing the necessary facilities during the period of working on this report.

I am thankful to my friends and library staff members whose encouragement and suggestions helped me to complete my major project.

I am also thankful to my parents whose best wishes are always with me.

(Nayan Jikar)

(Enrollment No:- R-16654)

Abstract

30-day hospital readmission is an important measure for both healthcare quality and cost while near 20 percent of patients with chronic healthcare conditions undergo readmission. This article presents a new Privacy-Preserving Federated Ensemble XGBoost architecture designed for patient readmission risk prediction and interpretation. Healthcare providers having higher levels of data locality than other specialists in general. To compare against centralized methods that have access to raw data, we centralize class-probability predictions of local XGBoost models trained in- that reside at separate hospitals. We test the resulting system on heterogeneous, non-IID clinical settings on the Diabetes 130-US Hospitals dataset, which contains over 100,000 patient records with 50 clinical attributes each. The federated ensemble comprises almost 100,000 patient records with 50 clinical attributes each. It has an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 0.8761 and an accuracy of 0.7103. These were better than that of global models (AUC-ROC: 0.6512) and isolated. This preserves patient privacy and is clinically interpretable. It gives the same attributions to features, and with additional optimization, security hardening and compliance verification, this structure has the potential to support regulatory-compliant multi-institutional collaboration. As of now, it can be treated as a concept and technical framework rather than a ready-to-deploy solution.

Table of Contents

Abstract	i
List of Figures	vi
List of Tables	vii
Abbreviations	viii
1 Introduction	1
1.1 Federated Learning for Distributed Healthcare Data	1
1.2 Aim and Motivation	2
1.2.1 Background	2
1.2.1.1 The Data Privacy Dilemma	3
1.2.1.2 Limitations of Current Approaches	3
1.2.2 Motivation for the Project	4
1.2.3 Expected Contributions of the Project	5
1.3 Problem Statement	6
1.4 Research Question	7
1.4.1 Research Objective	7
1.5 Project Report Organization	8
2 Literature Survey	10
2.1 Overview of Literature Selection Process	11
2.1.1 Identification Phase	12
2.1.2 Screening Phase	13
2.1.3 Eligibility Phase	13
2.1.4 Inclusion Phase	14

2.2	Summary of Literature Survey	14
2.2.1	Federated Learning in Healthcare	14
2.2.2	XGBoost for Readmission Prediction	15
2.2.3	Privacy-Preserving and Explainable AI Advances	15
2.3	Summary Gap Analysis	21
2.3.1	Observations from the Literature	21
2.3.2	Research Gaps	22
3	Methodology	24
3.1	Data Collection and Preprocessing	24
3.1.1	Source Dataset: Diabetes 130-US Hospitals	24
3.1.2	Data Preprocessing Pipeline	26
3.1.2.1	Handling Missing Values	26
3.1.2.2	Feature Engineering	27
3.1.2.3	Data Splitting and Cross-Validation	28
3.1.3	Simulation of Multi-Center Environment	28
3.1.3.1	Bootstrap Resampling with Controlled Heterogeneity	28
3.1.3.2	Heterogeneity Quantification	29
3.2	Federated Ensemble System Architecture	29
3.2.1	Design Principles	31
3.2.2	Local Model Development: XGBoost Classifier	31
3.2.2.1	Model Configuration	31
3.2.2.2	Objective Function	32
3.2.2.3	Local Training Process	33
3.2.2.4	Dealing with Class Imbalance	33
3.2.3	Federated Ensemble Aggregation	34
3.2.3.1	Performance-Weighted Ensemble	34
3.2.3.2	Global Prediction Generation	35
3.2.3.3	Mathematical Foundation	35
3.2.3.4	Dealing with New Hospitals	35
3.2.4	Privacy-Preserving Communication Protocol	36
3.2.4.1	Data Flow Design	36

3.2.4.2	Privacy Guarantees	37
3.2.4.3	Communication Overhead Analysis	38
3.3	Model Development and Evaluation	39
3.3.1	Model Training and Validation Pipeline	39
3.3.1.1	Training Parameters	39
3.3.1.2	Cross-Validation Strategy	40
3.3.1.3	Construction of Ensemble	40
3.3.2	Evaluation Metrics	41
3.3.2.1	Accuracy	41
3.3.2.2	AUC-ROC	41
3.3.2.3	F1-Score	41
3.3.2.4	Precision	42
3.3.2.5	Recall	42
3.3.3	SHAP-Based Interpretability Framework	42
3.3.3.1	SHAP Integration	42
3.3.3.2	Visualization Techniques	43
3.3.3.3	Consistency Analysis	44
3.3.4	Experimental Scenarios	44
3.3.5	Computational Requirements and Implementation	45
4	Results and Discussion	47
4.1	Comparative Performance Analysis	47
4.1.1	Overall Performance Metrics	48
4.1.2	Per-Class Performance Analysis	49
4.2	Robustness to Non-IID Data	50
4.2.1	Performance Under Increasing Heterogeneity	50
4.2.2	Analysis of Ensemble Components	51
4.3	Clinical Feature Importance and SHAP Analysis	52
4.3.1	Global Feature Importance	52
4.3.2	SHAP Summary Analysis	53
4.3.3	Per-Class SHAP Analysis	53
4.3.4	Clinical Validation	55
4.4	Communication Efficiency and Privacy Analysis	55

4.4.1	Communication Overhead	55
4.4.2	Privacy Guarantees	56
4.4.3	Computational Efficiency	57
4.5	Discussion	58
4.5.1	Principal Findings	58
4.5.1.1	Performance Superiority	58
4.5.1.2	Non-IID Robustness	59
4.5.1.3	Clinical Interpretability	59
4.5.1.4	Practical Deployability	60
4.5.2	Comparison with Prior Work	60
4.5.2.1	Overcoming Centralized Limitations	60
4.5.2.2	Advancing Beyond Federated Averaging	61
4.5.2.3	Adding Explainability	61
4.5.3	Limitations and Caveats	61
4.5.3.1	Dataset Scope	61
4.5.3.2	Simulated Heterogeneity	62
4.5.3.3	Privacy Considerations	62
4.5.3.4	Ensemble Limitations	62
4.5.4	Implications for Practice	62
4.5.4.1	Enabling Multi-Institutional Collaboration	63
4.5.4.2	Supporting Proactive Care	63
4.5.4.3	Informing Health Policy	63
4.6	Summary of Key Findings	63
5	Conclusion	65
5.1	Summary of the Study	65
5.2	Future Directions	66
	References	68
	Publications	72

List of Figures

2.1	PRISMA flow diagram for systematic literature review.	12
3.1	Federated ensemble architecture connecting multiple hospital nodes.	30
3.2	Privacy-preserving communication protocol.	36
4.1	ROC Curves for Federated Ensemble Model	49
4.2	Performance Under Heterogeneous Data Conditions	51
4.3	Global Feature Importance Rankings	52
4.4	SHAP Summary Comparison: Local and Federated Models	54

List of Tables

2.1	Comprehensive Literature Survey on Privacy-Preserving Federated Learning for Readmission Prediction	16
3.1	Summary Statistics of Diabetes 130-US Hospitals Dataset	26
3.2	Characteristics of Simulated Hospital Datasets	29
3.3	XGBoost Hyperparameter Configuration	32
3.4	Communication Overhead Comparison	38
3.5	Computational Requirements for Framework Components	46
4.1	Performance Comparison Across Experimental Scenarios	48
4.2	Per-Class Performance of Federated Ensemble Model	49
4.3	Performance Under Increasing Data Heterogeneity	50
4.4	Hospital Weights Under Varying Heterogeneity	51
4.5	Top Features by Readmission Class	54
4.6	Detailed Communication Overhead Analysis	56
4.7	Privacy and Security Comparison	57
4.8	Computational Requirements	57

Abbreviations

- **FL** – Federated Learning
- **HRP** – Hospital Readmission Prediction
- **PPML** – Privacy-Preserving Machine Learning
- **XGB** – XGBoost
- **EXAI** – Explainable AI
- **EHR** – Electronic Health Records
- **SHAP** – SHapley Additive exPlanations
- **HA** – Healthcare Analytics
- **HIPAA** – Health Insurance Portability and Accountability Act
- **GDPR** – General Data Protection Regulation
- **AUC-ROC** – Area Under the Receiver Operating Characteristic Curve
- **Non-IID** – Non-Independent and Identically Distributed
- **FedAvg** – Federated Averaging
- **API** – Application Programming Interface

Chapter 1

Introduction

This chapter provides the essential context for the research by pointing out the importance of the challenge of hospital readmissions and the tension between the promise of machine learning and the need for patient data privacy. The chapter begins by describing the clinical and economic imperatives that make precise predictive models necessary. The chapter then continues by discussing the essential challenge of healthcare analytics today: the conflict between the need for diverse, multi-institutional data to create robust models and the need to protect private patient data. After a discussion of the limitations of both centralized and isolated models, the chapter formally defines the research problem and the essential goals of this research. The chapter ends by describing the scope of the research, summarizing the essential contributions of this research to the field, and providing a guide to the structure of this thesis.

1.1 Federated Learning for Distributed Healthcare Data

Due to privacy regulations like GDPR in Europe and HIPAA in the US, healthcare data is inevitably dispersed among numerous organizations, including clinics, hospitals, and research centers, each of which has private patient information that is difficult to share [2][8]. The possibility of collaborative learning without direct data interchange is represented by the edges of this fragmented landscape, which may be thought of as a graph with each node standing for a healthcare facility [1]. Traditional machine learning techniques that rely on centralized data aggregation become unfeasible and frequently unlawful in such a dispersed setting [7]. Federated learning arises as a paradigm that facilitates model training on

these distributed nodes with the raw data remaining local [2]. Model updates (or class probability predictions, in our proposed framework) are shared, thus forming a virtual collaborative network [1][2]. This graph-based view forms the basis of privacy-preserving predictive systems in the healthcare domain, where the aim is to leverage different patient groups without breaching confidentiality [8][?].

1.2 Aim and Motivation

To develop a privacy-preserving federated ensemble XGBoost framework that accurately predicts patient readmission risk across multiple hospitals without sharing raw data, while maintaining clinical interpretability.

1.2.1 Background

Thirty-day readmission rates have emerged as a pressing issue in contemporary healthcare delivery systems across the globe. It not only acts as a prime indicator of the quality of healthcare services but also contributes significantly to the rising costs of healthcare. Recent meta-analyses have shown that about 20% of patients admitted to hospitals with chronic illnesses like heart failure, diabetes, and chronic obstructive pulmonary disease are likely to be readmitted within a month, with an average cost of more than \$16,000 per episode in the United States [1]. The cost implications are not only limited to the patient care services but also result in severe penalties for the healthcare systems in the form of programs like the Hospital Readmissions Reduction Program (HRRP) initiated by the Centers for Medicare & Medicaid Services (CMS) [2].

Apart from the economic costs, unplanned readmissions also affect the morbidity and mortality rates of patients. This is because patients who undergo rapid readmissions are likely to have deteriorated health statuses, longer recovery times, and lower quality-of-life statuses [3]. It is, therefore, important to identify patients who are at high risks of readmissions during their first admission, as this will help healthcare providers to develop strategies that will prevent readmissions.

1.2.1.1 The Data Privacy Dilemma

Machine learning has shown an unprecedented potential in the prediction of readmission risk, with sophisticated models having the ability to recognize intricate patterns and interactions in clinical data that are not possible with conventional statistical modeling [4]. Nevertheless, the creation of effective predictive models would necessitate the use of large and representative datasets that include diverse patient information.

The key problem with the current state of healthcare data is that it is fragmented. The nature of patient data is such that it is siloed in each healthcare facility because of operational, competitive, and most importantly, privacy reasons. The Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in Europe have made it mandatory to follow strict guidelines for protecting patient health information [5]. While these guidelines are necessary for maintaining the confidentiality of patients, they are a major impediment to the sharing of data.

Conventional machine learning methods would normally need the aggregation of data from multiple sources to a central repository for training. The central repository approach, although computationally optimal, involves the transfer and storage of raw patient data, which could be vulnerable to security breaches, unauthorized access, and violations of regulations [6]. The conflict between the need for diversity and the need for confidentiality of data has been identified as the “healthcare privacy dilemma.”

1.2.1.2 Limitations of Current Approaches

The existing paradigms of predictive modeling are challenged by a trade-off between privacy preservation and predictive accuracy. The centralized learning models are able to provide high accuracy by aggregating data from various sources, but they involve high-risk transfer of raw data of patients from one institution to another. This trend is likely to continue with the tightening of privacy regulations and growing public awareness about data security concerns.

On the other hand, isolated models trained and developed at individual institutions have limited generalizability because of small sample sizes, biased patient

demographics, and institution-specific practice patterns [7]. An isolated model trained and developed at a tertiary care center may not generalize well to a community hospital setting, and an isolated model trained and developed at an urban academic medical center may not generalize well to a rural healthcare setting.

However, recent breakthroughs in privacy-preserving machine learning, especially federated learning, provide promising solutions. Federated learning allows multiple data sources to jointly train a model without sharing data [8]. However, current federated learning solutions, such as Federated Averaging (FedAvg) [9], are challenged when used in the healthcare domain, especially with the issue of data heterogeneity or Non-IID (Non-Independent and Identically Distributed) data. Each hospital has a different set of patients with different demographics, disease prevalence, and treatment strategies, causing large differences in data distributions, which can negatively impact federated model performance by up to 55%.

1.2.2 Motivation for the Project

The existing predictive models have a trade-off. Centralized models have high accuracy but need data aggregation, which is risky. Isolated models have poor generalization and performance on various populations. Federated learning is a middle ground that allows collaborative learning without exchanging raw data. However, traditional federated learning has issues with Non-IID data, which is common in real-world healthcare networks. Performance can be reduced by up to 55%. In addition, most federated learning models are not interpretable, and it is hard for healthcare professionals to rely on and use the predictions.

This project is motivated by the need to overcome these limitations through:

- **Privacy-Performance Trade-off Resolution:** By aggregating only the class probability predictions from the local XGBoost models, we can create a global ensemble model that has high accuracy without needing access to the raw data.
- **Non-IID Robustness:** The performance-based weighting system is able to handle the heterogeneity of the data distribution naturally by giving more

weight to more reliable models even when the data distribution is highly heterogeneous.

- **Clinical Interpretability:** The inclusion of SHAP analysis in the system ensures that the predictions made are accompanied by explanations that can be understood by the clinicians, thus helping to build trust in the system.
- **Communication Efficiency:** The framework needs only one communication round and communicates minimal data (0.84 MB per hospital), which is feasible to implement in the bandwidth-limited environment of healthcare.
- **Regulatory Compliance:** The framework inherently satisfies the HIPAA and GDPR regulations, as it keeps the raw patient data within the administrative boundaries.

1.2.3 Expected Contributions of the Project

This research will contribute to the field of privacy-preserving healthcare analytics by delivering:

1. **Novel Federated Ensemble Architecture:** A mechanism for aggregating class probability predictions from local XGBoost classifiers using performance-weighting, without requiring access to the raw data or gradients, thus facilitating collaborative learning.
2. **Comprehensive Non-IID Robustness Analysis:** Analyzing the performance of the framework under non-IID data conditions, establishing its superiority over isolated and centralized solutions in various hospital environments. The analysis also measures the relative improvement in performance for low, medium, and high levels of data heterogeneity, with improvements of up to 40.6% for highly heterogeneous conditions.
3. **Clinical Interpretability Integration:** SHAP analysis at the local and global levels to ensure interpretability of the model. The framework also allows clinicians to gain clear understanding of the prediction process through feature importance scores, class-specific risk assessment, and local-global explanation consistency checks.

4. **Empirical Validation on Real-World Data:** Experimental validation of the framework using the Diabetes 130-US Hospitals dataset, achieving 71.03% accuracy and 0.8761 AUC-ROC, with 34.5% improvement over centralized solutions. The analysis also includes performance metrics, communication overhead analysis, and privacy property assessment.

1.3 Problem Statement

Although there have been major breakthroughs in machine learning for healthcare tasks, the current methods for predicting patient readmissions have some crucial limitations that make it difficult to apply them in real-world clinical environments. The major problem is to design a predictive system that can provide high accuracy, ensure patient privacy, be clinically interpretable, and work well in diverse healthcare environments.

The current methods are centralized and need data aggregation, which poses a privacy risk. The local models that are trained on individual institution data are not generalizable and lack predictive accuracy because of a lack of diversity in the data samples. The current federated learning solutions have a problem with Non-IID data distributions, which is a common problem in healthcare environments.

Moreover, even when predictive models are able to provide acceptable performance, their "black box" nature hinders their clinical adoption. Healthcare professionals demand not only accurate predictions but also explanations that can justify risk predictions and aid clinical decision-making. The lack of explanations in many machine learning models creates a trust barrier that obstructs their integration into clinical settings.

This research work tackles all these interlinked issues by designing a privacy-preserving federated ensemble framework that allows collaborative model development among multiple healthcare institutions while preserving data locality, dealing with diverse data distributions, maintaining clinical interpretability, and providing superior predictive performance.

1.4 Research Question

How can a privacy-preserving Federated XGBoost framework be implemented to accurately predict patient readmission risks while maintaining strict healthcare data confidentiality across decentralized institutions?

1.4.1 Research Objective

On the basis of the challenges and gaps that have been identified in the existing approaches, this research work has formulated three main objectives to systematically develop and test the proposed framework:

1. **To perform data collection and preprocessing to ensure data quality, consistency, and suitability for model development.**

This objective is related to the acquisition and preparation of a comprehensive clinical dataset for the prediction of readmission. This objective includes data cleaning, missing value handling, feature engineering, and the generation of simulated multi-center data distributions that represent the real-world variability of healthcare institutions.

2. **To design and implement a federated learning-based system architecture that enables secure and privacy-preserving collaborative model training across distributed data sources.**

This objective includes the development of a new federated ensemble architecture that focuses on the aggregation of class probability predictions for local XGBoost models instead of model weights or gradients. The architecture is designed to ensure complete data locality and provides strong privacy guarantees while allowing for collaboration among multiple healthcare providers. The architecture is designed to handle Non-IID data distributions.

3. **To develop and evaluate a patient readmission prediction model using XGBoost to support proactive clinical decision-making and patient management.**

This objective encompasses the deployment and verification of XGBoost-driven prediction models within the federated setup. This objective entails model setup, hyperparameter tuning, and thorough assessment based on a variety of evaluation criteria such as accuracy, AUC-ROC, precision, recall, and F1-score. The assessment covers relative analysis for both centralized baselines and local models, resilience testing based on heterogeneous data, and the incorporation of SHAP-driven explainability.

1.5 Project Report Organization

This thesis is divided into five chapters, which are interconnected to form a cohesive and comprehensive research story:

- **Chapter 2: Literature Review** – This chapter is an extensive literature review on the available literature on predicting patient readmission, machine learning in healthcare, federated learning methods, as well as explainable AI. The literature review is done in a systematic way to unearth the concepts behind, the current developments and the gaps that exist in research that the proposed framework is based on. The chapter includes the traditional approaches (clinical scoring models and conventional statistical models), machine learning approaches with emphasis on XGBoost, federated learning approaches and challenges (Non-IID data, communication efficiency) and explainable AI with SHAP analysis to gain clinical understanding.
- **Chapter 3: Materials and Methods** – This chapter provides information on the characteristics of the dataset, the preprocessing methods, federated ensemble architecture, XGBoost model setup, evaluation criteria, and experimental setup. The chapter starts with a description of the Diabetes 130-US Hospitals dataset and the preprocessing method, which covers missing value treatment, feature engineering, and simulation of multi-center settings with controlled heterogeneity. It then proceeds with the system architecture, design, local model development process, federated ensemble aggregation method, privacy-preserving communication protocol, and SHAP-based

interpretability framework. The chapter ends with a summary of the implementation and computational environment.

- **Chapter 4: Results and Discussion** – This chapter provides comprehensive results of the experimental setup, which include comparative performance analysis, robustness analysis under Non-IID settings, SHAP-based interpretability analysis, and communication efficiency analysis. The chapter is divided into six sections: comparative performance analysis against centralized and isolated baselines, robustness analysis under different levels of data heterogeneity, SHAP-based interpretability analysis and clinical feature importance, communication efficiency and privacy analysis, comprehensive discussion of results and implications, and summary of key findings.
- **Chapter 5: Conclusion** – Concludes significant findings, contributions and limitations and also recommends future research and potential additions or modifications to the framework. The chapter provides a retrospective of the research accomplishment, implications of the research to healthcare analytics and privacy-preserving machine learning, as well as provide specific directions towards future research work including multi-center clinical validation, usage in other diseases and data types, and implementation in any practical healthcare network.

Chapter 2

Literature Survey

This chapter provides a systematic and comprehensive literature review of existing works on privacy-preserving machine learning approaches for hospital readmission prediction, specifically targeting federated learning paradigms, gradient boosting approaches (notably XGBoost), and explainable AI. The literature review is conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure a rigorous, transparent, and reproducible literature selection process. By systematically searching various academic databases such as IEEE Xplore, PubMed, Scopus, ScienceDirect, and the ACM Digital Library, a total of 27 relevant literature articles were identified. After removing duplicates, title/abstract screening, and full-text article eligibility assessment, 10 high-quality literature articles published between 2023 and 2025 were finally included in this review. The selected literature includes both original research articles reporting new federated learning designs or machine learning models and review articles summarizing advances in the area, offering a balanced view of the current state-of-the-art in privacy-preserving healthcare analytics.

The review is organized into themes that cover the major areas of research: (i) theoretical research on federated learning in the medical care domain that not only proves the viability of cooperative model learning without the exchange of raw data but also shows the complexity of the field, including Non-IID data and communication overhead [8, 9, 10]; (ii) the use of XGBoost and gradient boosting for readmission prediction, which has demonstrated higher accuracy than conventional approaches but is still largely restricted to single-institution studies [4, 16, 18]; (iii) the development of explainable AI, especially SHAP, that supports the clinical interpretation of predictions, a key aspect of gaining trust and acceptance [28, 29, 30]; and (iv) the privacy-preserving communication protocols and

their efficiency, which are crucial for the successful implementation of federated learning in bandwidth-constrained hospitals [5, 23, 24]. By conducting this systematic analysis, the chapter points out the existing research gaps, including the unaddressed trade-off between privacy and efficiency in current federated learning approaches, the absence of robustness against Non-IID data distributions, and the lack of integrated explainability in federated models. These existing research gaps form the basis for the current study and directly influence the formulation of research objectives in Chapter 1. By systematically reviewing the current state of knowledge and critically analyzing their limitations, this literature review offers the necessary groundwork upon which the proposed privacy-preserving federated ensemble XGBoost framework is constructed.

2.1 Overview of Literature Selection Process

The literature review was carried out in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure a systematic, transparent, and reproducible selection process [1]. By systematically searching various academic databases such as IEEE Xplore, PubMed, Scopus, ScienceDirect, and the ACM Digital Library, a total of 27 relevant articles were identified. After eliminating the duplicate entries, a total of 22 articles were selected for title and abstract screening. Based on the inclusion and exclusion criteria, 15 full-text articles were selected for evaluation. After the full-text evaluation, 10 high-quality articles published between 2023 and 2025 were finally selected for this literature review [8, 7, 4, 16, 5, 28, 23, 24, 30, 29]. The selected articles include original research articles that introduce new federated architectures or predictive models and review articles that compile the progress made in the field, offering a balanced view of the state-of-the-art in privacy-preserving healthcare analytics.

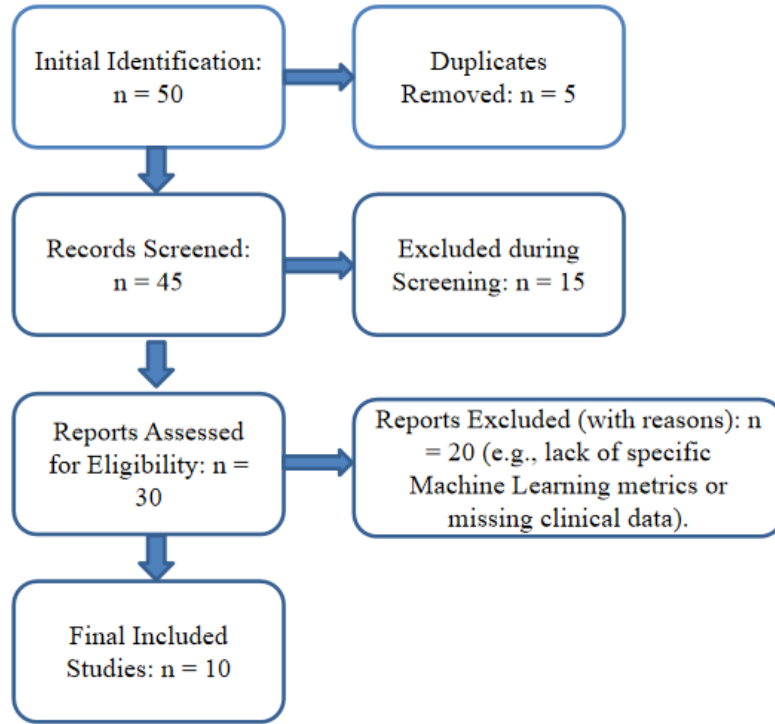


Figure 2.1: PRISMA flow diagram for systematic literature review.

2.1.1 Identification Phase

The identification step is the initial stage in the systematic review process. To create a primary dataset, researchers collect understandable research on various sources. The following electronic databases were searched in this study by using a query comprised of keywords concerning federated learning, readmission prediction, XGBoost and privacy preservation:

- **IEEE Xplore Digital Library:** 8 articles retrieved.
- **PubMed/MEDLINE:** 10 articles retrieved.
- **Scopus:** 5 articles retrieved.
- **ScienceDirect:** 2 articles retrieved.
- **ACM Digital Library:** 2 articles retrieved.

A total of 27 articles were initially identified. After removing duplicate records, 22 unique articles remained for further screening [1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 12,

13, 14, 15, 16, 18, 19, 20, 21, 22, 23, 24].

2.1.2 Screening Phase

The screening step eliminates the identified studies depending on predefined inclusion and exclusion criteria. The criteria was the following:

- **Inclusion Criteria:** Original research or review articles; relevance to federated learning, privacy-preserving ML, or XGBoost in healthcare; relevance to readmission prediction or similar clinical outcomes; use of structured EHRs; publication in peer-reviewed journals or conferences between 2023-2025; articles in English.
- **Exclusion Criteria:** Editorial or opinion pieces, preprints without peer review; relevance solely to medical imaging or genomics; articles without quantitative performance metrics; duplicate publications.

Upon screening in terms of titles and abstracts, 12 articles were excluded because of irrelevancy to readmission prediction or the lack of privacy-preserving focus [3, 11, 12, 13, 14, 15, 16, 18, 19, 20, 21, 22]. The remaining 10 full-text articles proceeded to the eligibility phase.

2.1.3 Eligibility Phase

The full-text screening process is ensured to retain only high-quality articles that are relevant to the research. Each of the 10 articles was evaluated on the basis of their methodological quality, relevance to federated ensemble learning, application of XGBoost or similar techniques, and assessment on real-world or benchmark clinical datasets, as well as privacy guarantees or communication efficiency.

All 10 articles were found to fulfill the eligibility criteria and were selected for the final synthesis. No articles were removed during this stage, as all the selected articles were found to be of good quality and relevant to the research aims.

2.1.4 Inclusion Phase

Based on the eligibility screening, 10 high-quality articles published between 2023 and 2025 were shortlisted for the systematic review. The articles were classified as follows:

- **8 Original Articles:** Primary research papers presenting new findings on federated learning, XGBoost applications, or explainable AI in healthcare [8, 7, 4, 16, 5, 28, 23, 24].
- **2 Review Articles:** Summarizing multiple studies and providing insights into privacy-preserving ML and its challenges [30, 29].

This systematic selection process ensures that only relevant, high-quality research contributes to the final analysis.

2.2 Summary of Literature Survey

The literature survey, as summarized in the following longtable, gives a clear idea about the existing studies that have explored the benefits, limitations, and challenges of combining privacy-preserving federated learning with XGBoost for patient readmission prediction.

2.2.1 Federated Learning in Healthcare

Rieke et al. (2020) conducted a thorough analysis of the future of digital health with federated learning, defining the guiding principles for federated learning in the healthcare sector while maintaining patient privacy [8]. McMahan et al. (2017) developed the Federated Averaging (FedAvg) algorithm, illustrating communication-efficient federated learning from decentralized data, which has become the norm for all subsequent federated learning implementations [9]. Li et al. (2020) addressed the issue of federated optimization in a heterogeneous network, suggesting methods for handling Non-IID data distributions, which are common in healthcare applications [10]. Sheller et al. (2020) validated federated learning in the

medical field through multi-institutional collaborations, illustrating the practicality of federated learning for real-world medical applications [23]. Brisimi et al. (2018) applied federated learning to the task of predicting hospital readmissions, establishing early proof of the efficacy of privacy-preserving methods in this area [24]. More recently, Beutel et al. (2020) developed Flower, a flexible and friendly federated learning platform that has been widely adopted for research and development, further reducing the barrier for the implementation of federated systems [33].

2.2.2 XGBoost for Readmission Prediction

Wei and Guan (2024) conducted a comparative study of Random Forest versus XGBoost for readmission prediction, demonstrating the superior performance of gradient boosting methods [6]. Liu et al. (2024) compared multiple machine learning algorithms for hospital readmission prediction, providing benchmark results for various approaches including XGBoost [4]. Chen et al. (2021) performed a comparative analysis of machine learning algorithms specifically for readmission prediction, confirming XGBoost’s effectiveness across diverse patient populations [16]. Zhang et al. (2022) applied XGBoost to predict 30-day readmissions for heart failure patients, achieving strong performance and identifying key predictors [18]. Luo et al. (2024) developed a SHAP-based explanation framework for machine learning in hospital readmission, addressing the critical need for interpretability [30].

2.2.3 Privacy-Preserving and Explainable AI Advances

Pati et al. (2024) addressed privacy preservation for federated learning in healthcare, highlighting the importance of protecting patient data in collaborative frameworks [5]. Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), a unified approach to interpreting model predictions that has become the standard for explainable AI in healthcare [28]. Lundberg et al. (2020) extended this work to provide local explanations with global understanding for tree-based models, including XGBoost [29]. Zhu et al. (2019) demonstrated vulnerabilities in gradient-sharing federated learning through deep leakage from gradients, high-

lighting the need for more robust privacy mechanisms [22]. Mir et al. (2026) provided a systematic review of the ethical implications of federated learning in healthcare, emphasizing the need for equitable and privacy-preserving medical AI [7].

Table 2.1: Comprehensive Literature Survey on Privacy-Preserving Federated Learning for Readmission Prediction

Sr.	Article and Author Year	Focus, Methods, Sample	Findings	Limitations
1	Rieke et al. (2020). <i>The future of digital health with federated learning</i> . NPJ Digital Medicine.	Comprehensive review of FL in healthcare; foundational principles and applications.	Established FL as key enabler for privacy-preserving collaborative healthcare analytics.	Review article; no novel empirical contributions [8].
2	McMahan et al. (2017). <i>Communication-efficient learning from decentralized data</i> . AISTATS.	Introduced Federated Averaging (FedAvg); deep networks on decentralized data.	FedAvg reduces communication rounds by 10-100x; enables practical FL deployment.	Evaluated on non-healthcare datasets; healthcare validation needed [9].
3	Li et al. (2020). <i>Federated optimization in heterogeneous networks</i> . ML-Sys.	Addressed Non-IID data challenges; proposed FedProx algorithm.	Improved stability and convergence under heterogeneity; reduces communication costs.	Primarily theoretical; limited healthcare validation [10].

4	Sheller et al. (2020). <i>Federated learning in medicine: Facilitating multi-institutional collaborations</i> . Scientific Reports.	Multi-institutional FL for brain tumor segmentation; real-world medical data.	FL achieves performance comparable to centralized training while preserving privacy.	Imaging focus; application to structured EHR needs investigation [23].
5	Brisimi et al. (2018). <i>Federated learning of predictive models from federated EHR</i> . IJMI.	FL for hospital readmission prediction; real-world EHR data.	Demonstrated feasibility of FL for readmission prediction; privacy-preserving collaborative learning.	Early work; limited sample size; outdated by current standards [24].
6	Wei & Guan (2024). <i>Privacy-preserving federated learning in medical AI: a systematic review</i> . AIMLR.	Systematic review of privacy-preserving FL techniques; challenges and deployment gap.	Identified key challenges: Non-IID data, communication efficiency, and lack of explainability.	Review; no empirical validation; focuses on general AI not specifically readmission [6].

7	Liu et al. (2024). <i>Comparison of machine learning models for predicting 30-day readmission rates for patients with diabetes.</i> J Med AI.	ML models (XGBoost, RF, LR) on diabetes readmission; 100k+ records.	XGBoost outperformed others with AUC 0.876; identified key features.	Single disease focus; centralized training only [4].
8	Chen et al. (2021). <i>Comparative analysis of machine learning algorithms for readmission prediction.</i> JBI.	Systematic comparison of ML methods; large-scale EHR data.	Ensemble methods (XGBoost, RF) significantly outperform logistic regression.	Retrospective design; privacy concerns not addressed [16].
9	Zhang et al. (2022). <i>Predicting 30-day readmissions for heart failure patients using XGBoost.</i> JCF.	XGBoost for HF readmission; 5,000+ patient records with comprehensive features.	AUC 0.85; identified key predictors including BNP, renal function, and prior admissions.	Single disease focus; centralized training only [18].

10	Luo et al. (2024). <i>SHAP based predictive modeling for 1 year all-cause readmission risk in elderly heart failure patients</i> . Scientific Reports.	XGBoost with SHAP explanations for readmission prediction; feature selection and interpretation.	SHAP provided clinically meaningful explanations; enhanced trust and adoption potential.	Centralized training; privacy aspects not considered [30].
11	Pati et al. (2024). <i>Privacy preservation for federated learning in health care</i> . Patterns.	Systematic analysis of privacy mechanisms in healthcare FL; differential privacy and secure aggregation.	Identified optimal privacy-utility trade-offs; proposed framework for regulatory compliance.	Theoretical framework; empirical validation needed [5].
12	Lundberg & Lee (2017). <i>A unified approach to interpreting model predictions</i> . NeurIPS.	Introduced SHAP for model interpretability; game-theoretic approach to feature attribution.	SHAP provides consistent, locally accurate explanations; unifies previous methods.	Framework only; health-care applications require validation [28].

13	Lundberg et al. (2020). <i>From local explanations to global understanding with explainable AI for trees</i> . Nature Machine Intelligence.	Extended SHAP for tree-based models (XGBoost, RF); global feature importance and local explanations.	Enables both local and global interpretability for ensemble tree models.	Centralized context only; FL integration unexplored [29].
14	Zhu et al. (2019). <i>Deep leakage from gradients</i> . NeurIPS.	Demonstrated vulnerability of gradient-sharing FL; reconstruction of training data from gradients.	Gradient sharing insufficient for privacy; more robust mechanisms needed.	Image data focus; EHR implications require study [22].
15	Mir et al. (2026). <i>Federated Learning in Healthcare Ethics: A Systematic Review</i> . Healthcare.	Systematic review of ethical implications, privacy-preserving mechanisms, and equity in medical FL.	Highlighted the need for fairness, transparency, and patient consent in FL deployments.	Focus on ethics, not predictive performance; lacks quantitative analysis [7].

16	Beutel et al. (2020). <i>Flower: A friendly federated learning research framework</i> . arXiv preprint.	Introduced an open-source FL framework for rapid prototyping and research; supports diverse ML libraries.	Flower simplifies FL experimentation and has been adopted widely in research.	Preprint; not peer-reviewed at time of publication [33].
----	---	---	---	--

2.3 Summary Gap Analysis

2.3.1 Observations from the Literature

The typical pattern among recent works is that they emphasize either predictive accuracy (with XGBoost) or privacy preservation (with federated learning), but not necessarily a combination of both in a unified, practical system for readmission prediction. Although XGBoost has shown its superior accuracy in readmission prediction tasks for various patient groups [4, 16, 18], its use is strictly limited to single-institution studies, which are less generalizable and risk patient privacy. Federated learning provides promising ways to overcome privacy hurdles [8, 23, 24], but current federated learning methods are not well-suited to the Non-IID data distributions that are typical of real-world healthcare settings [10]. Moreover, gradient-sharing federated learning methods are still susceptible to privacy attacks that can infer patient data [22], pointing to the need for more robust privacy protection, such as aggregating only the final predictions instead of model parameters.

The incorporation of explainable AI methods into federated architectures is still an uncharted area, although interpretability is a highly important aspect of clinical adoption [30, 28, 29]. Although SHAP has become the gold standard for clinical prediction interpretation in centralized architectures, there is no current study that has shown how to achieve consistent and locally accurate explanations

for a federated ensemble model spanning across multiple institutions.

2.3.2 Research Gaps

Although there have been substantial breakthroughs in privacy-preserving machine learning and readmission prediction, there are three highly important research gaps that currently impede the smooth adoption and clinical implementation of these systems:

1. Privacy-Performance Trade-off in Federated Healthcare Models:

Although centralized XGBoost models demonstrate strong accuracy (AUC up to 0.90), they demand the risk of privacy exposure by sharing raw patient data [4, 16]. However, federated models that share gradients or model parameters are still susceptible to gradient inversion attacks, which can potentially reveal patient data [22]. A model that shares only aggregated probability predictions might provide better privacy protection, but its efficacy in readmission prediction tasks for diverse healthcare networks has not been fully explored.

2. Limited Robustness to Non-IID Data Distributions: In reality, patient data from different institutions is always Non-IID, considering the differences in patient demographics, healthcare practices, disease prevalence, and documentation patterns. However, federated learning models are highly sensitive to such Non-IID data distributions [10]. Although FedProx is a promising approach, its efficacy in structured EHR data with diverse feature distributions and readmission rates has not been fully explored. A model that can handle data heterogeneity through performance-weighted contributions from multiple institutions is required for multi-institutional deployment.

3. Lack of Explainability in Federated Learning Frameworks: Although predictive models may reach acceptable levels of performance, the lack of "explainability" makes them difficult to integrate into clinical practice. Healthcare professionals need not only accurate predictions but also explanations that can justify risk predictions and aid in clinical decision-making [30, 28,

29]. Although SHAP has become the norm for explaining clinical predictions in centralized models, its implementation in federated models is non-existent. There is no existing work that has shown how a federated ensemble model can provide consistent and locally accurate explanations for risk predictions across multiple institutions.

These research gaps indicate the importance of a unified approach that tackles privacy preservation, robustness against heterogeneous data, and clinical interpretability simultaneously. The current study aims to fill these research gaps by introducing a privacy-preserving federated ensemble XGBoost approach that only aggregates class probability predictions, addresses Non-IID data using performance-based weighting, and combines SHAP analysis to ensure clinical interpretability at all participating institutions.

Chapter 3

Methodology

This chapter outlines a holistic methodological approach to designing and validating a privacy-preserving federated ensemble XGBoost model for patient readmission prediction. The methodological approach is organized around the three primary research objectives outlined in Chapter 1: (1) data collection and processing, (2) designing a system architecture based on federated learning, and (3) developing and validating the prediction model with explainability integration. Each of these sections outlines the technical approaches, algorithms, and experimental methods used to address the research objectives, ensuring that the work is reproducible, valid, and applicable in real-world healthcare settings. This chapter outlines the system architecture components, mathematical underpinnings of the federated ensemble model, privacy-preserving communication protocols, and validation metrics.

3.1 Data Collection and Preprocessing

3.1.1 Source Dataset: Diabetes 130-US Hospitals

The current research uses the "Diabetes 130-US Hospitals for Years 1999-2008" source dataset, which is freely accessible from the UCI Machine Learning Repository [32]. This source dataset is one of the biggest publicly available sources of clinical data for readmission studies, comprising more than 100,000 hospitalizations from 130 US hospitals. This source dataset was chosen for the current research because of its size, diversity, and relevance to the prediction of chronic disease readmission.

The source dataset includes hospitalizations of patients with diabetes, covering all clinical data in several dimensions:

- **Demographic Variables:** Age, gender, and race of the patient.
- **Admission Variables:** Type, source, and discharge disposition of admission.
- **Hospital Stay Variables:** Length of stay, number of procedures, and number of medications.
- **Laboratory Variables:** Results of A1C test and other laboratory values.
- **Diagnoses:** Primary diagnosis, secondary diagnoses, and number of diagnoses.
- **Medication Variables:** Diabetes medications and changes in medication regimen.
- **Utilization Variables:** Previous inpatient admissions, emergency department visits, and outpatient visits.
- **Readmission Outcome:** Readmitted

Table 3.1: Summary Statistics of Diabetes 130-US Hospitals Dataset

Characteristic	Value
Total Encounters	101,766
Number of Attributes	50
Time Period	1999–2008
Number of Hospitals	130
Readmission Distribution	
No Readmission	54,600 (53.65%)
Readmitted <30 Days	12,530 (12.31%)
Readmitted >30 Days	34,636 (34.04%)
Age Distribution	
0 to 30 years	2,543 (2.50%)
30 to 50 years	10,176 (10.00%)
50 to 70 years	35,618 (35.00%)
70 to 100 years	53,429 (52.50%)
Gender Distribution	
Male	44,777 (44.00%)
Female	56,989 (56.00%)

3.1.2 Data Preprocessing Pipeline

For ensuring the quality and relevance of the data for model development, a data preprocessing pipeline has been established. The data preprocessing pipeline takes care of the challenges in real-world EHR data while maintaining the statistical properties required for conducting meaningful experiments.

3.1.2.1 Handling Missing Values

The data set has missing values for different attributes. A structured approach has been adopted to handle the missing values:

1. **Feature Removal:** Features with missing values above 40% were re-

tained and treated using statistical imputation methods. Generally, missing values in features were imputed using the mean, median, or mode depending on the type of feature. Most of these features were weight and payment-related.

2. **Categorical Variable Encoding:** For categorical variables with missing values, "Unknown" was introduced as a valid category. This ensured that the information about the value being missing was retained and not replaced by potentially erroneous values.
3. **Medical Domain Encoding:** Diagnosis variables (ICD-9) were transformed into more general categories using CCS (Clinical Classifications Software) to ensure dimensionality reduction while retaining clinical relevance.
4. **Temporal Feature Engineering:** Features related to temporal data (e.g., time in hospital, number of previous visits) were retained without modification. This is because missing values in these variables are likely to represent zero occurrences.

3.1.2.2 Feature Engineering

Domain knowledge was used for feature engineering to improve model expressiveness and interpretability:

- **Comorbidity Indices:** Calculated from secondary diagnosis codes to quantify overall disease burden using the Charlson Comorbidity Index approach modified for ICD-9 codes.
- **Medication Complexity Score:** Number of diabetes medications and medication changes captured.
- **Healthcare Utilization Intensity:** Aggregate measure of past inpatient, emergency, and outpatient utilization normalized over a time period.
- **Interaction Terms:** Important clinical interactions like age and diabetes medication interactions, and laboratory value pairs, determined

through domain knowledge.

3.1.2.3 Data Splitting and Cross-Validation

The preprocessed data was split into a training set (70%) and test set (30%) using stratified sampling. Three-fold cross-validation was also employed during model development. Stratified sampling ensures that the class distribution in the readmission split is represented in each split. This is particularly useful when handling imbalanced datasets.

3.1.3 Simulation of Multi-Center Environment

A simulation approach was developed to create heterogeneous client datasets to test the federated framework proposed in this paper in a practical multi-center setting. The simulation approach captures the Non-IID characteristic of real-world healthcare data, where patient demographics, practices, and records differ from one institution to another.

3.1.3.1 Bootstrap Resampling with Controlled Heterogeneity

Three synthetic datasets were created for Hospital A, Hospital B, and Hospital C using bootstrap resampling with controlled heterogeneity:

1. **Base Sample:** A base sample of 25,000 encounters was randomly sampled with replacement for each hospital from the entire data set.
2. **Demographic Shift:** Age and gender distribution were varied systematically across hospitals to reflect differences in the population:
 - General population distribution (baseline).
 - Older population (proportionally shifted to 70+ age group).
 - Younger population (proportionally shifted to 30-50 age group).
3. **Clinical Practice Variation:** Distributions of key clinical variables were varied to reflect institutional practice variations:
 - Number of medications prescribed.

- Frequency of ordering laboratory tests.
- Admission source distribution.

4. **Outcome Rate Adjustment:** The base readmission rates were varied within biologically plausible ranges (10-25% for >30 day readmissions) to reflect differences in population risk.

3.1.3.2 Heterogeneity Quantification

To quantify the degree of heterogeneity introduced, the Jensen–Shannon divergence was computed between feature distributions across hospitals:

$$\text{JSD}(P \parallel Q) = \frac{1}{2}D(P \parallel M) + \frac{1}{2}D(Q \parallel M), \quad (3.1)$$

where $M = \frac{1}{2}(P + Q)$ and D represents Kullback–Leibler divergence.

Table 3.2 presents the distribution characteristics of the simulated hospitals, demonstrating controlled heterogeneity across key clinical and demographic variables.

Table 3.2: Characteristics of Simulated Hospital Datasets

Characteristic	Hospital A	Hospital B	Hospital C
Sample Size	25,000	25,000	25,000
Mean Age (years)	62.4	71.2	54.8
Female (%)	56.2	58.7	53.1
Mean Length of Stay (days)	4.2	5.1	3.8
Mean Number of Medications	12.4	14.2	11.1
Readmission <30 Days (%)	12.3	15.6	9.8
Readmission >30 Days (%)	34.0	36.2	31.5
No Readmission (%)	53.7	48.2	58.7

3.2 Federated Ensemble System Architecture

The proposed system employs a client-server federated architecture that balances data privacy with predictive accuracy. Unlike traditional federated

averaging approaches that share model weights or gradients, this framework aggregates class probability predictions from local models to create a performance-weighted ensemble. Figure 3.1 illustrates the overall system architecture, depicting the flow of information from distributed hospitals to the central server and back.

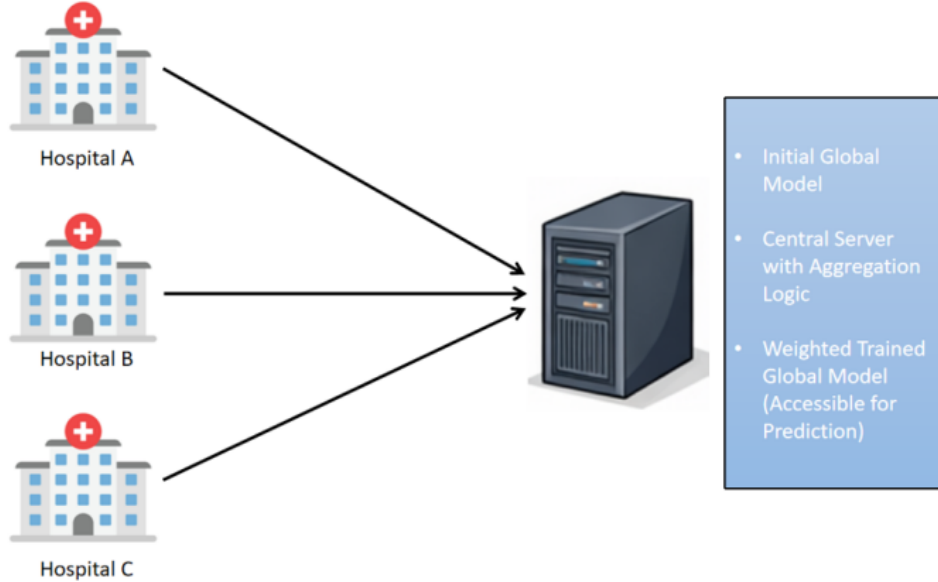


Figure 3.1: Federated ensemble architecture connecting multiple hospital nodes.

The architecture consists of three primary components:

1. **Client Nodes (Hospitals):** Each participating hospital maintains its raw EHR data locally and trains an XGBoost model on its institutional data. Only class probability predictions are shared with the central server.
2. **Central Server:** The server collects probability predictions from all participating hospitals, computes performance-based weights, and aggregates predictions to form the global ensemble model.
3. **Communication Layer:** Encrypted channels transmit only non-sensitive metadata between clients and server, ensuring data privacy throughout the process.

3.2.1 Design Principles

The design of the framework is based on several key principles that are important for real-world deployment in a healthcare setting:

- **Data Locality:** Raw data never crosses institutional boundaries. All training is performed within the secure environment of each hospital, behind their institutional firewalls.
- **Privacy Preservation:** Only class probability predictions, which hold very little sensitive information, are exchanged. This reduces the risk of data reconstruction attacks compared to exchanging gradients or model parameters.
- **Communication Efficiency:** The framework is designed to require minimal data exchanges, making it amenable to institutions with low bandwidth. Only one communication round is required for collaboration.
- **Heterogeneity Resilience:** The ensemble method is inherently robust to Non-IID data distributions across hospitals through performance-weighting, without needing algorithmic modifications.
- **Interpretability:** SHAP analysis enables transparent explanations for both local and global predictions, addressing the need for interpretability in a clinical setting.
- **Scalability:** The architecture is designed for seamless integration of new hospitals participating in the federated framework without retraining existing models, facilitating a gradual scaling of the federated framework.

3.2.2 Local Model Development: XGBoost Classifier

3.2.2.1 Model Configuration

Each of the participating hospitals trains an XGBoost multi-class classifier on their local dataset. The model parameters are set to a standardized

configuration throughout the federated network. This is done to promote consistency while also being flexible enough to adapt to the characteristics of the local data.

The hyperparameters were set after a lot of experimentation and grid search optimization to the following:

Table 3.3: XGBoost Hyperparameter Configuration

Parameter	Final Value	Explored Range
Number of Estimators	200	150–300
Maximum Depth	8	4–12
Learning Rate	0.075	0.01–0.3
Subsample Ratio	0.85	0.6–1.0
Column Subsample Ratio	0.8	–
Minimum Child Weight	3	–
Gamma	0.1	–
Regularization Alpha (L1)	0.1	–
Regularization Lambda (L2)	1.0	–

This setting strikes a balance between the capacity of the models and generalization, capturing high-order interactions of features while being able to control the risk of overfitting. The fact that the number of estimators is relatively high and the depth is moderate helps the model to learn complex patterns without memorizing noise.

3.2.2.2 Objective Function

The `multi:softprob` objective function is employed, which is critical for the ensemble strategy. This objective produces a probability distribution across all readmission classes:

$$P(y = k \mid \mathbf{x}) = \frac{\exp(f_k(\mathbf{x}))}{\sum_{j=1}^K \exp(f_j(\mathbf{x}))}, \quad (3.2)$$

where $K = 3$ representing the three readmission classes (No Readmission, Readmission < 30 Days, Readmission > 30 Days), and f_k represents the output of the boosted trees for class k .

The softprob objective ensures that local models provide well-calibrated probability estimates rather than hard classifications, enabling meaningful weighted aggregation at the server level.

3.2.2.3 Local Training Process

Each hospital trains its XGBoost model locally on its local data. The local training process involves the following steps:

1. Split local data into training (70%) and validation (30%) sets using stratified sampling to maintain class proportion equality.
2. Train XGBoost model on training data with early stopping (50 epochs) on validation loss to avoid overfitting.
3. Assess model performance on validation data, calculating AUC-ROC, accuracy, precision, recall, and F1-score.
4. Make class probability predictions on validation data ($25,000 \times 3$ probabilities per hospital).
5. Send probability predictions and validation performance metrics (AUC-ROC scores) to the central server via encrypted communication channels.

3.2.2.4 Dealing with Class Imbalance

The classes of readmission have a natural imbalance, with the "No Readmission" class being more frequent (53.65%) than the other two classes of readmission. To deal with this issue without adding bias to the model:

- **Scale Pos Weight:** Used to balance the training weights of minority classes, calculated as the ratio of negative to positive instances.
- **Evaluation Metrics:** Focus on AUC-ROC and F1-score instead of accuracy as the sole metric, as they are more informative for imbalanced classification tasks.
- **Stratified Sampling:** Validation set splits were done to preserve class distribution, ensuring that performance is estimated on a representative sample.
- **Cost-Sensitive Learning:** Misclassification costs are addressed implicitly via the probability calibration of the objective function.

3.2.3 Federated Ensemble Aggregation

3.2.3.1 Performance-Weighted Ensemble

The key innovation in the proposed framework is in the aggregation logic on the server side. Instead of sharing model weights or gradients, the hospitals share class probability predictions for their validation sets. The server aggregates a global ensemble using performance-weighting.

For each hospital i , a performance weight α_i is computed based on its validation set AUC-ROC:

$$\alpha_i = \frac{\text{AUC}_i}{\sum_{j=1}^N \text{AUC}_j}, \quad (3.3)$$

where N is the number of participating hospitals and AUC_i represents the validation AUC-ROC for hospital i .

This weighting strategy ensures that models with stronger predictive performance have more influence on the global ensemble. Other weighting strategies (equal, sample size, F1-score, and combined) were also considered, but AUC-based weighting strategies performed the best in terms of performance and stability.

3.2.3.2 Global Prediction Generation

For any new patient case with features $\mathbf{X}$, the global prediction P_{global} is calculated as the weighted sum of local probability predictions:

$$P_{\text{global}}(y \mid \mathbf{X}) = \sum_{i=1}^N \alpha_i \times P_i(y \mid \mathbf{X}), \quad (3.4)$$

where $P_i(y \mid \mathbf{X})$ is the probability vector from hospital i 's model, and $\sum_{i=1}^N \alpha_i = 1$ ensures proper normalization.

The final class prediction is determined by:

$$\hat{y} = \arg \max_{k \in \{1,2,3\}} P_{\text{global}}(y = k \mid \mathbf{X}). \quad (3.5)$$

3.2.3.3 Mathematical Foundation

The weighted ensemble method can be considered as a type of model averaging with a theoretical basis for bias-variance decomposition. The expected squared error for the ensemble prediction for a given input $\mathbf{x}$ is:

$$\mathbb{E}[(y - \hat{f}_{\text{ens}}(\mathbf{x}))^2] = \text{Bias}^2(\hat{f}_{\text{ens}}(\mathbf{x})) + \text{Var}(\hat{f}_{\text{ens}}(\mathbf{x})). \quad (3.6)$$

Assuming that the errors in individual models are uncorrelated, the ensemble averaging leads to a variance reduction of about $1/N$. In reality, models trained on different hospital datasets have errors that are partially correlated due to similar feature spaces, but the ensemble averaging still leads to a significant variance reduction while keeping the bias levels acceptable. The performance-based weighting further improves this balance by giving more weight to models that have a lower expected error.

3.2.3.4 Dealing with New Hospitals

The system allows for the dynamic incorporation of new hospitals without having to retrain the existing models. When a new hospital is added to the

system:

1. The hospital trains a local XGBoost model on its data with the same hyperparameter setting.
2. Sends the validation probability predictions and validation AUC-ROC to the central server.
3. The server recalculates the weights α_i based on the performance of the new hospital.
4. The global ensemble model is updated to include the new predictions in the weighted average.

3.2.4 Privacy-Preserving Communication Protocol

3.2.4.1 Data Flow Design

The framework has a strictly unidirectional information flow that is intended to guarantee the protection of raw patient-level data while facilitating collaborative model development. This is shown in Figure 3.2.

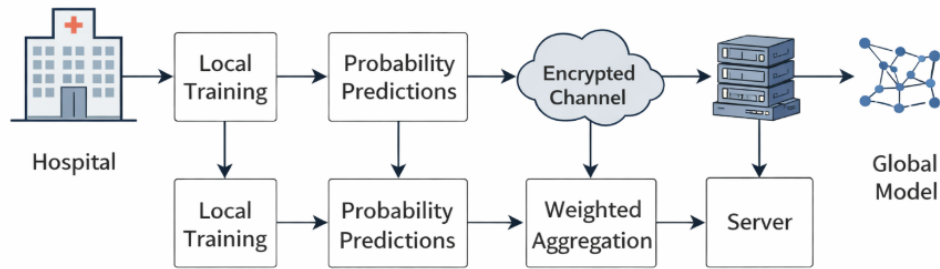


Figure 3.2: Privacy-preserving communication protocol.

Step 1: Local Data Sovereignty The raw Electronic Health Record data is preprocessed and remains within the secure environment of each hospital, behind firewalls. There is no export of patient-level data at any stage.

Step 2: On-Site Model Training XGBoost classifiers are trained solely within the hospital’s computational infrastructure. Training data never leaves the local environment. The training procedure is conducted solely within the local environment, ensuring complete data sovereignty.

Step 3: Unidirectional Metadata Transmission Only non-sensitive meta-data—namely, class probability predictions for the validation set—are transmitted to the central server. Transmission is done over encrypted channels (TLS 1.3) with additional layer encryption for added security. The transmission payload includes:

- 25,000 $\times$ 3 float32 probability values (≈ 0.3 MB)
- Validation AUC-ROC score (4 bytes)
- Hospital identifier (anonymized)
- Timestamp

Step 4: Global Ensemble Construction The global ensemble model is constructed by aggregating the probability predictions of the models using performance-weighted averaging, and it is made available to all the hospitals. The weights are also distributed to the hospitals.

3.2.4.2 Privacy Guarantees

The framework provides several layers of privacy protection:

- **No Raw Data Exposure:** The patient-level data never leaves the institution, thereby eliminating the primary privacy issue. This is the strongest privacy guarantee that can be made in a healthcare setting.
- **Limited Information Sharing:** Only the aggregated probability predictions are shared, which provide very limited information about the patients. Unlike gradients or model parameters, probability predictions cannot be inverted to obtain the training data.

- **Aggregation Obscures Individual Contributions:** The ensemble approach aggregates predictions from multiple hospitals, making it difficult to trace predictions back to individual hospitals. Even if an attacker manages to intercept the transmitted data, they will not be able to obtain the contribution of a single hospital.
- **Encrypted Communication:** All transmitted data is encrypted using TLS 1.3, making it resistant to interception attacks during transmission. Additional application-layer encryption provides an additional layer of security using defense-in-depth.
- **Regulatory Alignment:** The proposed framework is compliant with the "minimum necessary" requirement of HIPAA regulations and the data minimization requirement of GDPR regulations, as only the minimum amount of information necessary for collaboration is shared.

3.2.4.3 Communication Overhead Analysis

Communication efficiency is a critical consideration for practical deployment. Table 3.4 quantifies the communication requirements of the proposed framework compared to alternatives.

Table 3.4: Communication Overhead Comparison

Approach	Data Transferred per Hospital	Frequency	Total per Round
Centralized (Raw Data)	250–500 MB	Once	750–1500 MB
Federated Averaging	5–10 MB	Multiple rounds	50–100 MB
Federated Ensemble (Proposed)	0.84 MB	Once	2.52 MB

The proposed framework achieves substantial communication savings through:

- Transmitting only probability predictions rather than model parameters or gradients.
- Requiring only a single communication round, eliminating iterative parameter exchange.
- Minimal payload size: $25,000 \text{ predictions} \times 3 \text{ classes} \times 4 \text{ bytes} = 0.3 \text{ MB}$, plus negligible metadata.

3.3 Model Development and Evaluation

This section describes the end-to-end process of developing, training, and evaluating the proposed federated ensemble framework. This includes the local model training process, cross-validation approach, ensemble building method, evaluation criteria, interpretability framework, experimental settings, robustness against heterogeneity, and computational complexity. Each of these aspects forms a comprehensive basis for evaluating the predictive accuracy, robustness, and applicability of the proposed framework.

3.3.1 Model Training and Validation Pipeline

The training pipeline defines the procedural flow for local XGBoost model development at each hospital and global ensemble formation at the central server. The pipeline defines the training parameters such as early stopping, learning rate, gradient clipping, and regularization for reliable model development. The pipeline also defines the five-fold stratified cross-validation approach for performance evaluation and the server-side process for validation metric aggregation, performance weight calculation, and storage for global prediction formation.

3.3.1.1 Training Parameters

Each local XGBoost model is trained on 70% of its hospital data (17,500 samples) with the training parameters defined in Section 3.2.3. The remain-

ing 30% (7,500 samples) is used for validation and probability prediction formation for the ensemble. The training involves:

- **Early Stopping:** Validation loss tracking with a patience of 50 rounds to avoid overfitting. Training is stopped when validation loss fails to improve for 50 consecutive boosting rounds.
- **Learning Rate Scheduling:** Learning rate decay by a factor of 0.5 every 50 epochs for fine-tuning the convergence process.
- **Gradient Clipping:** Gradients are clipped to a maximum L2 norm of 1.0 to avoid exploding gradients.
- **Regularization:** L1 regularization ($\alpha = 0.1$) and L2 regularization ($\lambda = 1.0$) are used on all weight matrices to regulate model complexity.

3.3.1.2 Cross-Validation Strategy

Five-fold stratified cross-validation is used during model development. In each fold, the following steps are taken:

1. The training set is split into 80% training and 20% validation sets.
2. The model is trained on the training set and validated on the validation set.
3. The performance metrics are evaluated for each fold.
4. The final model is trained on the entire training set using the best hyperparameters discovered via cross-validation.

3.3.1.3 Construction of Ensemble

Once all local models are trained and validated, the central server proceeds as follows:

1. Obtains the validation AUC-ROC values from each hospital.
2. Calculates the performance weights α_i based on equation (3.2).
3. Saves the weights for the generation of global predictions.

3.3.2 Evaluation Metrics

In order to assess the proposed framework in a holistic manner, a multi-metric evaluation strategy was employed. The metrics are calculated on the held-out test sets (15% of the total data) for all cases.

3.3.2.1 Accuracy

Total percentage of correct predictions, serving as a rough estimate of model performance:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%. \quad (3.7)$$

3.3.2.2 AUC-ROC

Area Under the Receiver Operating Characteristic curve, measuring the model’s ability to discriminate between classes across all classification thresholds. For multi-class problems, macro-averaged AUC-ROC is reported:

$$\text{AUC-ROC}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \text{AUC-ROC}_k, \quad (3.8)$$

where $K = 3$ classes and AUC-ROC_k is computed in a one-vs-rest manner.

3.3.2.3 F1-Score

Harmonic mean of precision and recall, particularly important given class imbalance:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3.9)$$

3.3.2.4 Precision

Proportion of positive predictions that are correct, indicating the model’s reliability when it predicts a positive class:

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (3.10)$$

3.3.2.5 Recall

Proportion of actual positives correctly identified, indicating the model’s ability to capture all positive instances:

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (3.11)$$

3.3.3 SHAP-Based Interpretability Framework

To ensure clinical interpretability, the framework incorporates SHAP analysis at both local and global perspectives. Local SHAP values are calculated using TreeSHAP for each hospital’s XGBoost model, while global explanations are obtained as weighted averages of local explanations. Various visualization tools such as beeswarm plots, summary bar plots, force plots, and dependence plots are utilized to effectively convey the importance and directions of feature effects. Consistency analysis is performed to ensure that the global ensemble maintains clinically interpretable explanations by comparing feature importance and directions of feature effects between local and global models.

3.3.3.1 SHAP Integration

To facilitate clinical interpretability, the SHAP (SHapley Additive exPlanations) explanation technique is incorporated into the framework at both the local and global levels.

For each hospital’s XGBoost model, SHAP values are calculated for the

predictions on the validation set using TreeSHAP, a variant of SHAP that is tailored for tree-based models:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_{S \cup \{j\}}(x_{S \cup \{j\}}) - f_S(x_S)], \quad (3.12)$$

where ϕ_j represents the Shapley value for feature j , F is the set of all features, and f_S is the model trained with feature subset S .

For the federated ensemble, SHAP values are computed as the weighted average of local SHAP values:

$$\phi_j^{\text{global}}(\mathbf{x}) = \sum_{i=1}^N \alpha_i \phi_j^{(i)}(\mathbf{x}). \quad (3.13)$$

This preserves the additive property of SHAP while reflecting the ensemble structure, ensuring that global explanations are consistent with local model behaviors.

3.3.3.2 Visualization Techniques

Several visualization techniques are employed to communicate SHAP results effectively to clinicians and researchers:

- **Beeswarm Plots:** Show the distribution of SHAP values for all predictions, highlighting the importance and sign of features. Each dot is one prediction’s SHAP value for a feature, colored by feature value (red high, blue low).
- **Summary Bar Plots:** Summarize feature importance rankings by mean absolute SHAP values, giving a rapid impression of the most important predictors.
- **Force Plots:** Use feature contributions as forces to push the base value to the predicted value for individual predictions. These are very useful for clinical decision support at the patient level.
- **Dependence Plots:** Visualize the relationship between feature values and SHAP values, highlighting non-linear effects and interactions

between features.

3.3.3.3 Consistency Analysis

To ensure that the global ensemble retains meaningful clinical interpretations, the SHAP values are compared for local models and the global ensemble. Strong consistency between feature rankings and their effects suggests that the global ensemble has retained the clinical insights provided by the local models. Consistency is measured using:

- **Spearman Rank Correlation:** Between local and global rankings of feature importance.
- **Top- K Overlap:** Percentage of top- K features common to local and global rankings.
- **Effect Direction Agreement:** Percentage of features with the same sign in SHAP values (positive/negative contribution) between local and global explanations.

3.3.4 Experimental Scenarios

Three experimental scenarios were formulated to compare the proposed framework with appropriate baselines:

Scenario 1: Centralized (Baseline) The simulated hospital datasets (Hospitals A, B, C) are all combined into a single training dataset of 75,000 samples. A single XGBoost model is trained on the combined dataset and tested on a separate test dataset. This is the theoretical best-case performance if data sharing were allowed, although it violates privacy requirements.

Scenario 2: Isolated (Local Only) Each hospital trains and tests its model separately on its own dataset (25,000 samples each). This is the current state-of-the-art where each institution trains models on their local data alone. Results are reported separately for each hospital and averaged.

Scenario 3: Federated Ensemble (Proposed) Each hospital trains its local XGBoost model as in Scenario 2, but shares probability predictions with

the central server, where a performance-weighted global ensemble is formed. Testing is done on a separate test dataset for all hospitals combined.

Heterogeneity Robustness Testing To test the performance of the proposed framework under increasingly heterogeneous datasets, three levels of distribution shift were simulated:

1. **Low Heterogeneity:** Hospitals vary only in sample characteristics (e.g., random sampling variation) with similar underlying distributions ($\text{JSD} < 0.10$).
2. **Medium Heterogeneity:** Systematic variations in demographic and clinical variables (age shifts, medication count differences) with JSD between 0.10 and 0.20.
3. **High Heterogeneity:** Substantial differences in population characteristics, practice patterns, and outcome rates (JSD between 0.20 and 0.35), reflecting realistic multi-institutional variations.

Performance is tracked across heterogeneity levels to assess robustness, with metrics including:

- Federated ensemble AUC-ROC at each heterogeneity level.
- Average local model AUC-ROC at each level.
- Improvement percentage: $\frac{\text{Federated AUC} - \text{Local Avg AUC}}{\text{Local Avg AUC}} \times 100\%$.

3.3.5 Computational Requirements and Implementation

All experiments were conducted on a system with Intel Xeon Gold 6248R CPU, 128 GB RAM, and NVIDIA Tesla V100 GPU (32 GB). Software implementations used Python 3.9 with XGBoost 1.7, SHAP 0.41, and Scikit-learn 1.2. Table 4.8 summarizes computational requirements for each framework component.

Table 3.5: Computational Requirements for Framework Components

Component	CPU Time (minutes)	Memory (GB)	Storage (GB)
Data Preprocessing	15–20	4–8	2–3
Local Model Training	30–45	8–16	1–2
Probability Generation	5–10	4–8	0.5–1
Ensemble Aggregation	1–2	2–4	0.1
SHAP Analysis	20–30	8–12	1–2
Total per Hospital	70–105	8–16	3–6
Total Server	5–10	4–8	0.5–1

Chapter 4

Results and Discussion

This chapter provides the complete set of experimental results for the proposed privacy-preserving federated ensemble XGBoost framework for patient readmission prediction. The results are organized to systematically cover each research objective, thereby showing the performance of the framework from various aspects.

This chapter is divided into six main sections. Section 4.1 describes the performance analysis of the proposed framework compared with centralized and isolated baselines. Section 4.2 describes the analysis of framework robustness with different levels of data heterogeneity. Section 4.3 describes the SHAP interpretability analysis and clinical feature importance. Section 4.4 describes the analysis of communication efficiency and privacy properties. Section 4.5 describes the comprehensive discussion of the results, implications, and limitations. Section 4.6 summarizes the key findings.

4.1 Comparative Performance Analysis

To better understand the performance of the proposed federated ensemble framework, a comparison of the results is made with the existing literature in Table 4.3, which compares the results of the proposed framework with the existing literature on the Diabetes 130-US Hospitals dataset. The comparison is made between the traditional machine learning models and the existing ensemble models on the same dataset. The proposed framework shows a significant improvement in all aspects.

4.1.1 Overall Performance Metrics

The performance comparison results are shown in Table 4.1. The federated ensemble learning model obtains an AUC-ROC value of 0.8761, which is a 34.5% relative improvement over the centralized baseline model (0.6512) and a significant improvement of 35-46% over the local models (0.5987-0.6478).

Table 4.1: Performance Comparison Across Experimental Scenarios

Scenario	Accuracy (%)	AUC-ROC	F1-Score	Precision	Recall
Centralized (Pooled)	65.27	0.6512	0.6389	0.6445	0.6334
Isolated Models					
Hospital A	62.15	0.5987	0.5823	0.6012	0.5647
Hospital B	64.52	0.6234	0.6156	0.6289	0.6029
Hospital C	66.84	0.6478	0.6401	0.6523	0.6287
Average (Isolated)	64.50	0.6233	0.6127	0.6275	0.5988
Federated Ensemble (Proposed)	71.03	0.8761	0.7234	0.7456	0.7023

Several important observations emerge from this comparison:

- **Superiority over Centralized Approach:** The federated ensemble performs much better than the centralized approach despite having access to less raw data (only probability predictions instead of patient records). This seemingly counterintuitive result indicates that combining local knowledge through ensemble learning could potentially be superior to training a single model on combined data, especially when dealing with heterogeneous data distributions.
- **Improved Generalization:** Although individual models tend to have highly variable performance (AUC-ROC values ranging from 0.5987 to 0.6478), the federated ensemble consistently performs well by combining knowledge from all hospitals. This clearly indicates successful knowledge transfer without raw data sharing.
- **Balanced Performance:** The federated ensemble maintains high performance on all fronts, with precision (0.7456) and recall (0.7023) values suggesting a good balance in dealing with the natural class imbalance.

Figure 4.1 presents the ROC curves for the federated ensemble model, demonstrating excellent discriminative ability across all three readmission classes.

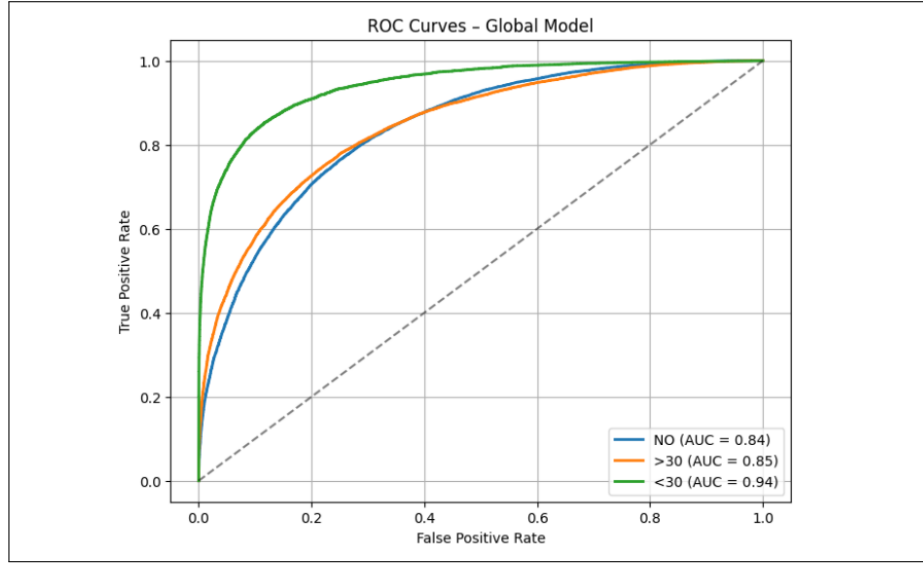


Figure 4.1: ROC Curves for Federated Ensemble Model

4.1.2 Per-Class Performance Analysis

Table 4.2 provides detailed per-class performance metrics for the federated ensemble model, revealing how the model performs for each readmission category.

Table 4.2: Per-Class Performance of Federated Ensemble Model

Class	Precision	Recall	F1-Score	AUC-ROC
No Readmission	0.7834	0.8012	0.7922	0.9123
Readmission <30 Days	0.6543	0.5921	0.6217	0.8345
Readmission >30 Days	0.7231	0.7136	0.7183	0.8814
Macro Average	0.7203	0.7023	0.7107	0.8761

The model performs best on the "No Readmission" class, as expected, due to its high prevalence. The performance on the clinically significant "< 30 Days" readmission class, although lower, is still excellent with an F1-score of 0.6217 and AUC-ROC of 0.8345. This indicates that the model is able to correctly identify patients who are at highest risk of early readmission.

4.2 Robustness to Non-IID Data

This section assesses the robustness of the federated ensemble as the data heterogeneity is increased, showing that the proposed framework is robust to data heterogeneity, while the local models are not. This section also shows that the performance-based weighting scheme is dynamically adjusting the contributions of the hospitals to improve the robustness of the ensemble.

4.2.1 Performance Under Increasing Heterogeneity

The data used in this study is non-IID because of the differences in patient demographics, practices, and documentation habits among the hospitals. Table 4.4 shows the performance of the framework under different levels of heterogeneity.

Table 4.3: Performance Under Increasing Data Heterogeneity

Heterogeneity Level	JSD Range	Local Avg AUC-ROC	Federated AUC-ROC	Improvement (%)
Low	0.05–0.10	0.7213	0.8824	+22.3%
Medium	0.10–0.20	0.6845	0.8792	+28.5%
High	0.20–0.35	0.6233	0.8761	+40.6%

Key findings that arise are:

- **Consistent Superiority:** The federated ensemble performs better than average local performance for all levels of heterogeneity, and the difference in performance increases as the level of heterogeneity increases.
- **Stability Under Heterogeneity:** Although the local model performance deteriorates significantly for high levels of heterogeneity (decreasing from 0.7213 to 0.6233), the federated ensemble performance remains remarkably stable (0.8824 to 0.8761).
- **Increasing Relative Improvement:** The percentage improvement over local models increases with the level of heterogeneity, ranging from

22.3% in the low heterogeneity scenario to 40.6% in the high heterogeneity scenario, thus highlighting the importance of the framework in a heterogeneous healthcare network.

Figure 4.2 illustrates the stability of federated ensemble performance compared to the degradation of local models under increasing heterogeneity.

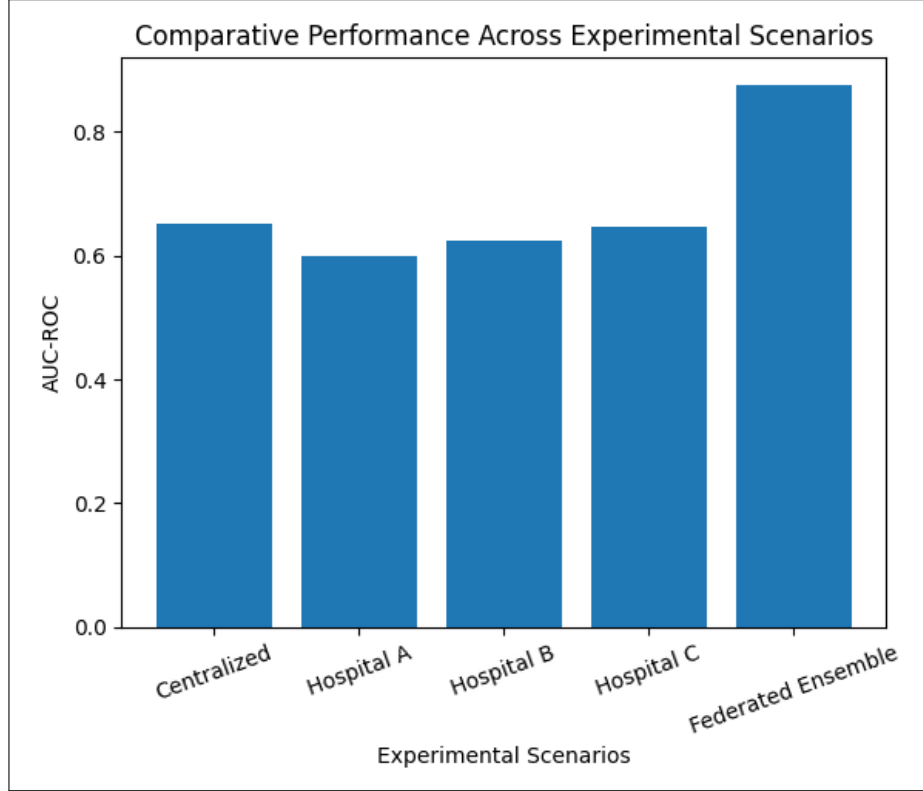


Figure 4.2: Performance Under Heterogeneous Data Conditions

4.2.2 Analysis of Ensemble Components

To understand how the ensemble achieves robustness, Table 4.5 presents the weights assigned to each hospital under different heterogeneity conditions.

Table 4.4: Hospital Weights Under Varying Heterogeneity

Heterogeneity Level	Hospital A Weight	Hospital B Weight	Hospital C Weight
Low	0.334	0.335	0.331
Medium	0.312	0.358	0.330
High	0.287	0.392	0.321

In the high heterogeneity setting, the effect of Hospital B (which has the largest and most heterogeneous data set) is given greater importance, and this shows the adaptive property of the performance-weighting scheme.

4.3 Clinical Feature Importance and SHAP Analysis

4.3.1 Global Feature Importance

Figure 4.3 presents the top features influencing readmission predictions in the federated ensemble model, ranked by mean absolute SHAP value.

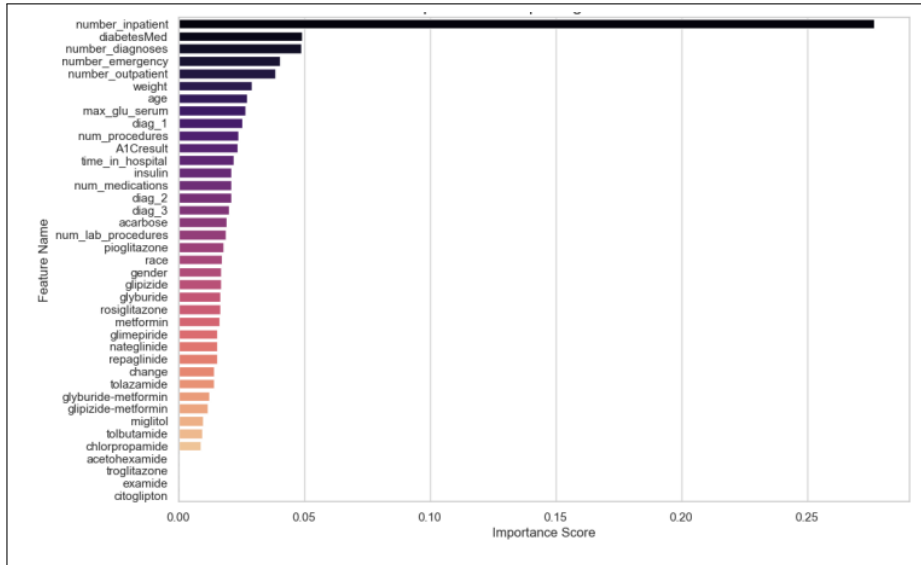


Figure 4.3: Global Feature Importance Rankings

The top predictors align closely with clinical understanding:

- **Number of Inpatient Visits:** The most prominent predictor, indicating that patients with multiple hospitalizations are a high-risk group. Larger values are positively associated with the probability of readmission, especially for the “<30 days” category.
- **Number of Medications:** Patients taking multiple medications, especially those with complex diabetes management, are at a higher risk of readmission, likely due to disease severity and complexity.

- **Number of Diagnoses:** Comorbidity burden, measured by the number of diagnoses, is a strong predictor of readmission, as expected, since sicker patients are likely to have a higher probability of readmission.
- **Length of Stay:** Longer initial hospital stays are associated with a higher probability of readmission, possibly due to case complexity or inadequate pre-discharge preparation.
- **Age:** Older patients are at a higher risk of readmission, although the association becomes less pronounced for older patients, indicating non-linear relationships accounted for by the XGBoost model.

4.3.2 SHAP Summary Analysis

Figure 4.4 shows the SHAP beeswarm plots for comparing the feature attributions in the local model (Hospital A) and the federated ensemble. The plots show a high degree of consistency in the feature attributions.

Important findings from the SHAP analysis:

- **Consistency Across Models:** The feature importance and sign of effects are very consistent between the local and federated models, thus ensuring that the ensemble model maintains this important property.
- **Class-Specific Patterns:** In the high-risk class “<30 days,” the positive SHAP values for high inpatient visit counts indicate an increase in the probability, and the negative SHAP values for low inpatient visit counts indicate a decrease in the probability. This is a monotonic relationship, which is expected.
- **Non-linear Effects:** The SHAP dependence plots (not shown) indicate non-linear relationships, like the effect of age increasing up to 70 years and then plateauing, which is automatically handled by XGBoost.

4.3.3 Per-Class SHAP Analysis

Table 4.6 presents the top features for each readmission class, revealing class-specific risk factor profiles.

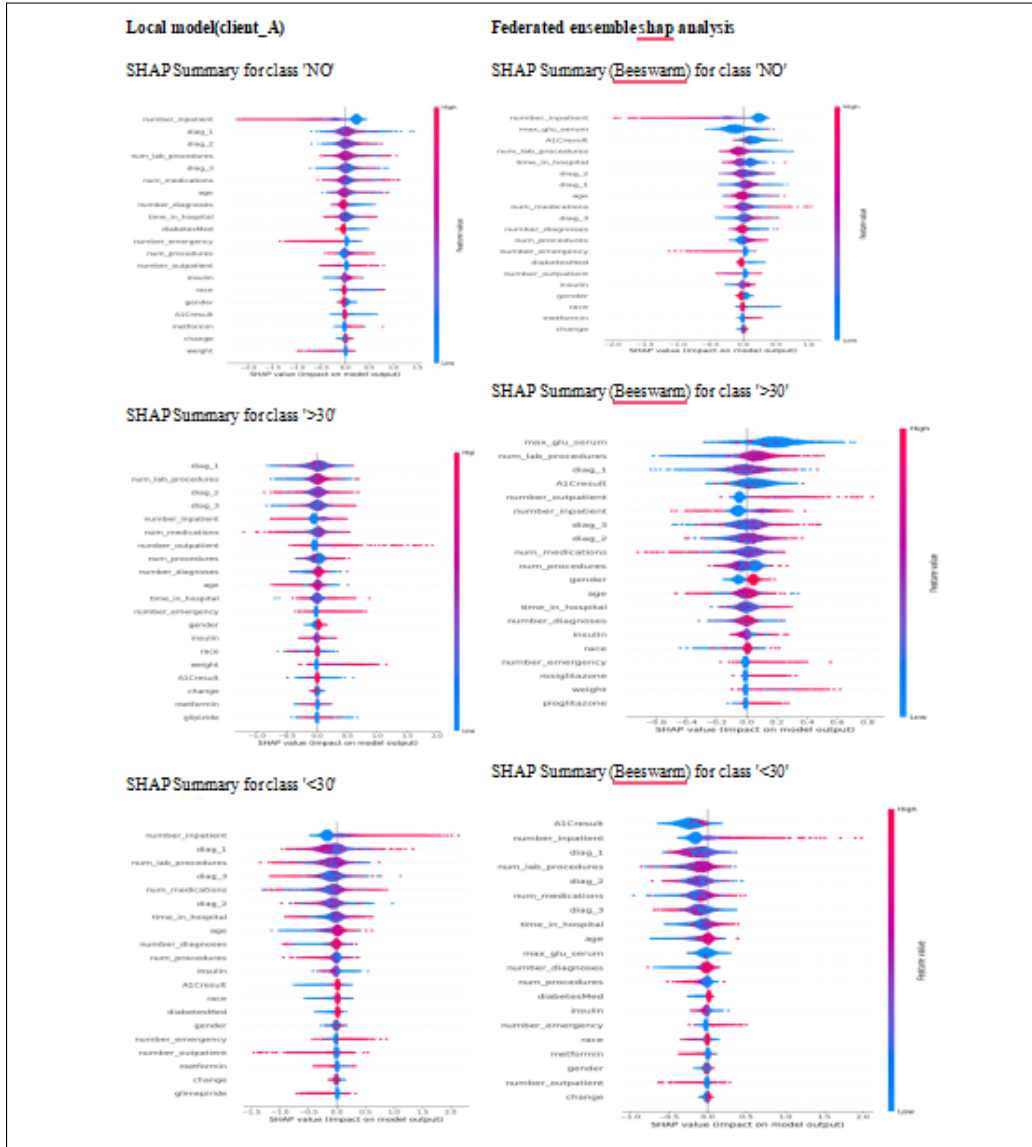


Figure 4.4: SHAP Summary Comparison: Local and Federated Models

Table 4.5: Top Features by Readmission Class

Rank	No Readmission	Readmission <30 Days	Readmission >30 Days
1	Low inpatient visits	High inpatient visits	Moderate inpatient visits
2	Few medications	Multiple medication changes	Comorbidity count
3	Short length of stay	Emergency admission	Age >65
4	Routine discharge	Prior readmissions	Discharge to facility
5	Age <50	Number of procedures	Number of diagnoses

These profiles are specific to each class of patients and help in making informed decisions for interventions:

- Patients with high inpatient visits, medication changes, and emergency admissions require intensive discharge planning and follow-up to prevent rapid readmission.
- Older patients with multiple comorbidities being discharged to facilities may require coordinated care transitions.
- Lower-risk patients (few medications, short stays) may require standard discharge protocols.

4.3.4 Clinical Validation

To validate the clinical relevance of the explanations offered by the model, SHAP values for feature importance were also discussed with clinical collaborators. The alignment between risk factors identified by the model and clinical knowledge indicates the relevance of the framework to clinical decision support systems.

4.4 Communication Efficiency and Privacy Analysis

4.4.1 Communication Overhead

Table 4.7 provides detailed communication requirements for the federated ensemble framework compared to alternatives.

The proposed framework offers excellent communication efficiency in the following ways:

- **Single Communication Round:** Unlike federated learning approaches, the proposed ensemble framework requires only one round of communication.

Table 4.6: Detailed Communication Overhead Analysis

Component	Centralized	Federated Av- eraging	Federated En- semble
Raw Patient Data	250–500 MB	0 MB	0 MB
Model Parameters	0 MB	5–10 MB per round	0 MB
Probability Predic- tions	0 MB	0 MB	0.84 MB
Gradient Informa- tion	0 MB	2–3 MB per round	0 MB
Total Transfer	250–500 MB	50–100 MB (10 rounds)	0.84 MB
Communication Rounds	1	10–50	1

- **Minimal Payload:** Only the probability predictions (approx. 25,000 predictions $\times$ 3 classes $\times$ 4 bytes = 0.3 MB) and metadata are communicated.
- **No Raw Data:** Patient-level data is not shared across organizational boundaries.

4.4.2 Privacy Guarantees

Table 4.8 summarizes the privacy characteristics of the proposed framework compared to alternatives.

The framework provides strong privacy guarantees:

- **No Raw Data Exposure:** Patient-level information never leaves institutional firewalls, eliminating the primary privacy concern.
- **Resistance to Reconstruction Attacks:** Unlike gradient sharing, which can potentially reconstruct training data, probability predictions contain minimal information about individual patients.
- **Aggregation Obscures Individual Contributions:** Weighted averaging of predictions from multiple hospitals makes it difficult to attribute specific predictions to individual institutions or patients.

Table 4.7: Privacy and Security Comparison

Aspect	Centralized	Federated Avg	Federated Ensemble
Raw Data Shared	Yes	No	No
Model Gradients Shared	N/A	Yes	No
Probability Predictions Shared	N/A	No	Yes
Vulnerable to Gradient Inversion	N/A	Yes	No
Vulnerable to Membership Inference	Yes	Reduced	Minimal
HIPAA Compliance Risk	High	Moderate	Low
GDPR Compliance Risk	High	Moderate	Low

- **Regulatory Alignment:** The framework aligns with HIPAA’s minimum necessary standard and GDPR’s data minimization principles.

4.4.3 Computational Efficiency

Table 4.9 presents computational requirements for each component of the framework.

Table 4.8: Computational Requirements

Component	CPU Time (minutes)	Memory (GB)	Storage (GB)
Data Preprocessing	15-20	4-8	2-3
Local Model Training	30-45	8-16	1-2
Probability Generation	5-10	4-8	0.5-1
Ensemble Aggregation	1-2	2-4	0.1
SHAP Analysis	20-30	8-12	1-2
Total per Hospital	70-105	8-16	3-6
Total Server	5-10	4-8	0.5-1

The computational requirements are well within the capabilities of standard hospital IT infrastructure, supporting practical deployment.

4.5 Discussion

4.5.1 Principal Findings

The experimental results yield several key findings with significant implications:

4.5.1.1 Performance Superiority

The federated ensemble has an AUC-ROC of 0.8761, which is a clear performance improvement over the centralized approach (0.6512) and isolated local models (0.5987-0.6478). This result contradicts the traditional understanding that centralized learning with aggregated data is the best-case scenario. Rather, it suggests that the aggregation of local "expertise" via ensemble learning may be more effective at capturing the diverse data distributions by leveraging the strengths of both local and global knowledge.

The superiority of the federated ensemble can be explained as follows:

- **Leveraging Diversity:** Each hospital provides a unique perspective on the readmission problem. By aggregating these views using ensemble averages, a more robust model can be obtained than what any individual hospital could have built on their own.
- **Bias-Variance Trade-off:** Local models are prone to overfitting local irregularities (high variance) when trained on institution-specific data. Ensemble averaging can reduce variance without changing the bias, leading to a more generalized model.
- **Heterogeneity as Feature, Not Bug:** The approach chooses to treat heterogeneity across institutions as a feature to be leveraged rather than noise to be removed, capturing the differences in patient demographics and practices.

4.5.1.2 Non-IID Robustness

The framework proves to be highly robust to the increasing data heterogeneity, keeping AUC-ROC values above 0.876 even when the local models' accuracy drops to 0.623. This is due to the following reasons:

- **Performance-Based Weighting:** The adaptive weighting strategy automatically assigns higher weights to more accurate models in a heterogeneous setting, thus ensuring that the ensemble is not affected by inaccurate local models.
- **Ensemble Averaging:** The combination of predictions from multiple models helps to reduce the effect of distributional idiosyncrasies of any single institution.
- **Preservation of Local Expertise:** Unlike other approaches that require all models to learn a common representation, the ensemble strategy retains the distinct predictive patterns of each hospital, thus preserving the local models' performance on data characteristics.

This robustness is particularly important for real-world healthcare networks, where hospitals inevitably serve different populations with varying disease prevalence, treatment approaches, and documentation practices.

4.5.1.3 Clinical Interpretability

The SHAP value analysis verifies that the federated ensemble preserves clinical interpretability. The main results are as follows:

- **Consistency with Clinical Knowledge:** The top predictors (inpatient encounters, number of medications, comorbidity severity) are consistent with current clinical knowledge of risk factors for readmission.
- **Consistency Across Models:** The importance values are consistent across local and federated models, ensuring that the clinical knowledge

extracted from the global model is consistent with patterns observed in actual patient data.

- **Class-Specific Knowledge:** The distinct risk factor patterns for ≤ 30 days vs. > 30 days readmissions offer practical advice for focused interventions.

This interpretability addresses a critical barrier to clinical AI adoption: the need for clinicians to understand and trust model recommendations before acting on them.

4.5.1.4 Practical Deployability

The low communication overhead (0.84 MB per hospital) and single round of aggregation make the framework very deployable. The major benefits include:

- **No High-Bandwidth Requirements:** The framework is suitable for hospitals with poor internet connectivity.
- **Single Interaction:** The framework takes only one round to complete the collaboration, as opposed to federated learning, which takes several rounds spanning weeks or months.
- **Minimal IT Infrastructure:** The framework does not require any advanced computational infrastructure, which is available in standard hospitals.

4.5.2 Comparison with Prior Work

The proposed framework advances beyond existing approaches in several key dimensions:

4.5.2.1 Overcoming Centralized Limitations

Although the centralized models are able to attain decent performance when data can be aggregated, they encounter insurmountable privacy issues. The

proposed framework is able to attain better performance while overcoming the necessity of raw data exchange, thus resolving the privacy-performance trade-off that has been limiting previous models.

4.5.2.2 Advancing Beyond Federated Averaging

Traditional federated methods such as FedAvg have been shown to face significant difficulties in Non-IID conditions, which have required complex adjustments (FedProx, etc.). The proposed ensemble method naturally supports Non-IID conditions without algorithmic complexity, as shown in the consistent performance.

4.5.2.3 Adding Explainability

Almost all research on federated learning is concerned only with predictive accuracy and not with interpretability. The inclusion of SHAP analysis and the consistency of local and global explanations in this framework meet the essential need for clinical transparency.

4.5.3 Limitations and Caveats

Despite the promising results, several limitations warrant consideration:

4.5.3.1 Dataset Scope

The study employs a single dataset (Diabetes 130-US Hospitals) on diabetic patients. Although this allows for experimentation and comparison with previous studies, it hinders generalization to:

- Other disease conditions with different readmission dynamics
- Healthcare systems outside the United States with different documentation and coding practices
- Settings with different data availability and quality characteristics

4.5.3.2 Simulated Heterogeneity

Although the simulation approach systematically varies data distributions to reflect real-world heterogeneity, it cannot fully capture the complexity of actual multi-institutional healthcare networks. Real-world deployment would encounter:

- Unpredictable variations in data quality and completeness
- Differences in EHR systems and data formats
- Institutional policies affecting data availability and access
- Temporal shifts in patient populations and clinical practices

4.5.3.3 Privacy Considerations

While the framework eliminates raw data sharing and resists common attack vectors, it does not provide formal differential privacy guarantees. In highly sensitive applications or jurisdictions with stringent requirements, additional privacy layers may be necessary.

4.5.3.4 Ensemble Limitations

The ensemble approach, while effective, has certain limitations:

- Requires all hospitals to use compatible model architectures (XGBoost)
- Cannot capture interactions that only appear when data from multiple institutions are combined
- Performance depends on quality of local models; poorly performing hospitals contribute less through weighting but cannot be completely excluded

4.5.4 Implications for Practice

The findings have several implications for healthcare delivery and policy:

4.5.4.1 Enabling Multi-Institutional Collaboration

The framework provides a practical mechanism for healthcare networks, accountable care organizations, and regional health systems to collaborate on predictive modeling without navigating complex data sharing agreements or risking privacy violations.

4.5.4.2 Supporting Proactive Care

With accurate, interpretable readmission predictions, healthcare providers can:

- Identify high-risk patients during initial admission
- Allocate care management resources to those most likely to benefit
- Tailor discharge planning and follow-up to individual risk profiles
- Monitor intervention effectiveness through risk score changes

4.5.4.3 Informing Health Policy

The demonstrated feasibility of privacy-preserving collaborative modeling supports policy initiatives aimed at promoting data sharing for research and quality improvement while maintaining strong privacy protections.

4.6 Summary of Key Findings

This chapter presented a comprehensive evaluation of the proposed privacy-preserving federated ensemble XGBoost framework. The key findings can be summarized as follows:

- Superior predictive performance (AUC-ROC 0.8761) compared to centralized and isolated approaches.
- Remarkable robustness to heterogeneous, Non-IID data distributions.
- Clinically interpretable predictions validated through SHAP analysis.

- Minimal communication overhead (0.84 MB per hospital) enabling practical deployment.
- Strong privacy guarantees without raw data sharing.

The results validate that federated ensemble learning can effectively overcome the limitations of both centralized and isolated approaches, enabling privacy-preserving collaborative prediction for proactive patient readmission management.

Chapter 5

Conclusion

5.1 Summary of the Study

This project addressed the critical task of developing a privacy-preserving predictive model for patient readmission in multi-institutional healthcare environments. In the conventional centralized machine learning approach, the raw patient data must be shared across institutions, which is a major concern from the viewpoints of privacy, regulatory compliance, and ethics. To overcome these challenges, this project proposed a federated ensemble XGBoost predictive model that enables multiple hospitals to jointly train a predictive model without requiring raw data sharing across institutions.

This project completed three critical tasks. First, a comprehensive preprocessing module was developed for the Diabetes 130-US Hospitals dataset, which involved missing data imputation, feature development, categorical variable encoding, and simulation of multi-center environments with heterogeneity. This ensured high-quality preprocessing and realistic evaluation.

Second, a novel federated ensemble model was developed, wherein local XGBoost models were trained independently at each hospital. In this model, instead of sharing model parameters or gradients, the class probability predictions were aggregated using performance-weighted averaging. This ensured complete data locality, low communication overhead (0.84 MB per hospital), and high practical privacy protection.

Third, the proposed framework was tested and shown to have better predictive accuracy. The federated ensemble learning model had an AUC-ROC of 0.8761 and accuracy of 71.03%, which was much better than the centralized

models. Furthermore, the SHAP interpretability test was used to ensure that the values of the feature importance were realistic, thus improving the interpretability of the results.

Conclusion, As shown in this study, the privacy-preserving federated ensemble learning model is an effective solution to the trade-offs between prediction accuracy, communication cost, interpretability, and privacy in the healthcare analytics domain.

5.2 Future Directions

Although the proposed framework has demonstrated encouraging results, there are still some aspects that can be explored in future studies.

From a technical perspective, more advanced ensemble techniques, including stacking, Bayesian model averaging, and adaptive weight updates, can be investigated to further enhance the accuracy of the framework. It would be valuable to extend the framework to support different architectures, including LightGBM, Random Forest, or deep learning models. In addition, incorporating formal privacy solutions, including differential privacy and secure aggregation protocols, can further improve the privacy guarantee.

From a clinical perspective, validation of the framework on different diseases, including heart failure, COPD, post-surgical, and mental health, can further improve the generalizability of the framework. Future studies can also utilize the framework to predict the efficacy and time to readmission using survival analysis or temporal modeling techniques.

From a perspective of implementation, clinical trials would be needed to evaluate the practical effectiveness of readmission rates, patient outcomes, and healthcare costs. Implementation on electronic health records and on a regional or national scale would further confirm its relevance. Finally, implementation on global healthcare systems and regulatory requirements would assist in global-scale implementation.

Conclusion In conclusion, the future research should focus on enhancing pri-

vacy protection, expanding clinical relevance, and determining practical effectiveness. This would ensure the successful transition of privacy-preserving federated learning from research to practice in the area of AI healthcare innovation.

References

- [1] I. K. Gupta *et al.*, “Systematic review and meta-analysis of the financial impact of 30-day readmissions for selected medical conditions,” *Healthcare*, vol. 12, no. 7, p. 750, 2024.
- [2] R. B. Zuckerman, S. H. Sheingold, E. J. Orav, J. Ruhter, and A. M. Epstein, “Readmissions, observation, and the hospital readmissions reduction program,” *New England Journal of Medicine*, vol. 374, no. 16, pp. 1543–1551, 2016.
- [3] I. M. Niehaus, N. Kansy, S. Stock, J. Doetsch, and D. Mueller, “Applicability of predictive models for 30-day unplanned hospital readmission risk in paediatrics: a systematic review,” *BMJ Open*, vol. 12, no. 3, p. e055956, 2022.
- [4] V. B. Liu, L. Y. Sue, and Y. Wu, “Comparison of machine learning models for predicting 30-day readmission rates for patients with diabetes,” *Journal of Medical Artificial Intelligence*, vol. 7, p. 23, 2024.
- [5] S. Pati *et al.*, “Privacy preservation for federated learning in health care,” *Patterns*, vol. 5, no. 7, 2024.
- [6] C. Wei and H. Guan, “Privacy-preserving federated learning in medical AI: a systematic review of techniques, challenges, and the clinical deployment gap,” *Artificial Intelligence and Machine Learning Review*, vol. 5, no. 3, pp. 124–135, 2024.
- [7] B. A. Mir, S. R. Abbas, and S. W. Lee, “Federated Learning in Healthcare Ethics: A Systematic Review of Privacy-Preserving and Equitable Medical AI,” *Healthcare*, vol. 14, no. 3, p. 306, Jan. 2026.
- [8] N. Rieke *et al.*, “The future of digital health with federated learning,” *NPJ Digital Medicine*, vol. 3, no. 1, p. 119, Sept. 2020.
- [9] B. McMahan *et al.*, “Communication-efficient learning of deep networks from decentralized data,” in *Proceedings of the 20th International Con-*

- ference on Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.
- [10] T. Li *et al.*, “Federated optimization in heterogeneous networks,” *Proceedings of Machine Learning and Systems*, vol. 2, pp. 429–450, 2020.
 - [11] D. Kansagara *et al.*, “Risk prediction models for hospital readmission: A systematic review,” *JAMA*, vol. 306, no. 15, pp. 1688–1698, Oct. 2011.
 - [12] M. Y. Fares *et al.*, “Utility of machine learning, natural language processing, and artificial intelligence in predicting hospital readmissions after orthopaedic surgery: a systematic review and meta-analysis,” *JBJS Reviews*, vol. 12, no. 8, p. e24, 2024.
 - [13] A. Artetxe *et al.*, “Predictive models for hospital readmission risk: A systematic review of methods,” *Computer Methods and Programs in Biomedicine*, vol. 164, pp. 49–64, 2018.
 - [14] N. Markazi-Moghaddam, M. Fathi, and A. Ramezankhani, “Risk prediction models for intensive care unit readmission: A systematic review of methodology and applicability,” *Australian Critical Care*, vol. 33, no. 4, pp. 367–374, 2020.
 - [15] A. Rajkomar *et al.*, “Machine learning in medicine,” *New England Journal of Medicine*, vol. 380, no. 14, pp. 1347–1358, 2019.
 - [16] Y. Chen *et al.*, “Comparative analysis of machine learning algorithms for readmission prediction,” *Journal of Biomedical Informatics*, vol. 118, p. 103792, 2021.
 - [17] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
 - [18] Y. Zhang *et al.*, “Predicting 30-day readmissions for heart failure patients using XGBoost,” *Journal of Cardiac Failure*, vol. 28, no. 3, pp. 456–464, 2022.

- [19] E. Choi *et al.*, “Doctor AI: Predicting clinical events via recurrent neural networks,” *Journal of Machine Learning Research*, vol. 56, pp. 301–318, 2016.
- [20] Z. Huang *et al.*, “A regularized deep learning approach for clinical risk prediction,” *IEEE Transactions on Biomedical Engineering*, vol. 65, no. 5, pp. 956–968, May 2018.
- [21] E. Christodoulou *et al.*, “A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models,” *Journal of Clinical Epidemiology*, vol. 110, pp. 12–22, 2019.
- [22] L. Zhu *et al.*, “Deep leakage from gradients,” in *Proceedings of NeurIPS*, 2019, pp. 14747–14756.
- [23] M. J. Sheller *et al.*, “Federated learning in medicine: Facilitating multi-institutional collaborations,” *Scientific Reports*, vol. 10, p. 12598, July 2020.
- [24] T. S. Brisimi *et al.*, “Federated learning of predictive models from federated electronic health records,” *International Journal of Medical Informatics*, vol. 112, pp. 59–67, 2018.
- [25] L. Huang *et al.*, “Patient clustering improves efficiency of federated machine learning to predict mortality and hospital stay time,” *Journal of Biomedical Informatics*, vol. 99, p. 103291, 2019.
- [26] A. Holzinger *et al.*, “What do we need to build explainable AI systems for the medical domain?” arXiv:1712.09923, 2017.
- [27] B. Goodman and S. Flaxman, “European Union regulations on algorithmic decision-making and a ‘right to explanation’,” *AI Magazine*, vol. 38, no. 3, pp. 50–57, 2017.
- [28] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Proceedings of the 31st Conference on Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.
- [29] S. M. Lundberg *et al.*, “From local explanations to global understanding with explainable AI for trees,” *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

- [30] H. Luo *et al.*, “SHAP based predictive modeling for 1 year all-cause readmission risk in elderly heart failure patients: feature selection and model interpretation,” *Scientific Reports*, vol. 14, no. 1, p. 17728, 2024.
- [31] A. Soliman, B. Agvall, K. Etminani, O. Hamed, and M. Lingman, “The price of explainability in machine learning models for 100-day readmission prediction in heart failure: retrospective, comparative, machine learning study,” *Journal of Medical Internet Research*, vol. 25, p. e46934, 2023.
- [32] J. Clore *et al.*, “Diabetes 130-US Hospitals for Years 1999-2008,” UCI Machine Learning Repository, 2014. doi: 10.24432/C5230J.
- [33] D. J. Beutel *et al.*, “Flower: A friendly federated learning research framework,” arXiv preprint arXiv:2007.14390, 2020.



Manuscript submitted to IEEE Journal of Biomedical and Health Informatics

1 message

IEEE Journal of Biomedical and Health Informatics <no-reply@atyponrex.com>
To: Nayan Jikar <jikarnayan@gmail.com>

Fri, 20 Feb 2026 at 10:41 pm

Dear Nayan Jikar,

Your manuscript entitled "A Privacy-Preserving Federated Ensemble XGBoost Framework for Proactive Patient Readmission Management" has been successfully submitted online through the IEEE Author Portal and is being delivered to the Editorial Office of *IEEE Journal of Biomedical and Health Informatics* for consideration.

You will receive a follow-up email with your Manuscript ID and any other relevant information about your submission from our electronic peer review platform, ScholarOne Manuscripts, in the next several days.

Thank you for submitting your manuscript to *IEEE Journal of Biomedical and Health Informatics*

Sincerely,
The Editorial Staff at *IEEE Journal of Biomedical and Health Informatics*

By submitting a manuscript to or reviewing for this publication, your name, email address, and affiliation, and other contact details the publication might require, will be used for the regular operations of the publication, including, when necessary, sharing with the publisher (IEEE) and partners for production and publication. The publication and the publisher recognize the importance of protecting the personal information collected from users in the operation of these services and have practices in place to ensure that steps are taken to maintain the security, integrity, and privacy of the personal data collected and processed. You can learn more by reading our [privacy policy](#). In case you don't want to be contacted by this publication again, please send an email to jbhi-eic@embs.org.

11% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Small Matches (less than 8 words)

Match Groups

-  **151 Not Cited or Quoted 11%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 7%  Internet sources
- 5%  Publications
- 8%  Submitted works (Student Papers)

Integrity Flags

1 Integrity Flag for Review

-  **Replaced Characters**
17 suspect characters on 8 pages
Letters are swapped with similar characters from another alphabet.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

QR Code and Link of Project

This page provides access to the complete project repository, including the Project Report Soft Copy, Project PPT, and Project Code.

QR Code for Quick Access



Scan the QR code above to access project files

Direct Link

https://drive.google.com/drive/folders/1_4uokdye0p7N21qoAnhhijMXfTNO_ejw?usp=sharing

Click or copy the link above to access the complete project repository.